2024 Philippine Clinical Practice Guidelines on the Diagnosis and Management of Acute Severe Blood Pressure Elevation

DISCLAIMER

This guideline is intended for use by general practitioners, allied health professionals, emergency medical personnel, specialists, and any healthcare workers who may encounter adult patients with hypertension. Although this guideline's recommendations represent a comprehensive review of the literature on the topic and the consensus of experts in the field, these should not be the sole consideration made by clinicians when caring for individual patients presenting with elevations in blood pressure. While guidelines provide a general framework for standards of care, clinicians must exercise their own clinical judgment for each case and consider each patient's goals, values, and preferences in the decision-making process.

Payors and policymakers, including hospital administrators and employers, may use this guideline as part of the considerations for developing policies and procedures, but this document should not be the sole basis for approval or denial of financial assistance and insurance claims. Recommendations from this guideline should also not be the only basis for any legal action.

The Task Force acknowledges the ongoing paradigm shift in hypertension which advocates the retirement of the concepts of "hypertensive urgency" and "hypertensive crisis." This movement is supported by the lack of evidence showing that acutely treating these conditions has any impact on the incidence of cardiovascular events, beyond that already afforded by antihypertensive therapy for chronic hypertension. Despite this, these terms are still widely used in local practice and medical literature. Hence, to ensure a comprehensive review of relevant literature and to aid the understanding of the audience, the Task Force decided to use the terms "hypertensive urgency" and "hypertensive crisis" in the formulation of clinical questions and search strategies.

The contents of this guideline are limited to the best available evidence as of the time of this writing. Aspects of diagnosis and management not studied in current literature cannot be fully addressed by this document and are considered in the context of expert opinion only.

Contact Us

Send us an email at phihypn@yahoo.com for any questions or clarifications regarding the outputs and processes of this Hypertension Emergency CPG.

TABLE OF CONTENTS

ABBREVIATIONS AND ACRONYMS

ABPM - ambulatory blood pressure monitoring
ACEi – angiotensin-converting enzyme inhibitor
ACEP – American College of Emergency Physicians
ACS – acute coronary syndrome
ACC – American College of Cardiology
ASCVD – atherosclerotic cardiovascular disease
AHA – American Heart Association
ARB - angiotensin-receptor blocker
BB – beta-blocker
BIHS – British and Irish Hypertension Society
BP - blood pressure
CAD - coronary artery disease
CCB – calcium channel blocker
CI - confidence interval
COI – conflict of interest
CP - consensus panel
CPG - clinical practice guideline
CV - cardiovascular
CVD - cardiovascular disease
DALY - disability-adjusted life year
DBP – diastolic blood pressure
DOH - Department of Health
ECG – electrocardiogram
ERE – evidence review expert
ESC - European Society of Cardiology
EtD – evidence to decision framework
GRADE - Grading of Recommendations, Assessment, Development, and Evaluation
HF – heart failure
HMOD – hypertension-mediated organ damage
ICH – intracranial hemorrhage
IV – intravenous
IHD – ischemic heart disease
ISH – International Society of Hypertension
LOS – length of stay
MACE – major adverse cardiovascular event
MI – myocardial infarction
NPV – negative predictive value
NSW – North South Wales
PPV – positive predictive value
PSH – Philippine Society of Hypertension
RCT – randomized controlled trial
SBP – systolic blood pressures
SC – Steering Committee
Sn – sensitivity
Sp – specificit

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HOW TO USE THIS DOCUMENT

This CPG contains a set of recommendations and best practice statements guided by systematic reviews of evidence and consensus panel discussions and intended to address eleven clinical questions on the diagnosis and management of severe BP elevation.

An overview of the goals and focus of this guideline can be found in the Introduction. The methodology implemented when creating this guideline can be reviewed on page 13. A summary of the recommendations and best practice statements can be found on page 17 and it is followed by evidence summaries to support the recommendation for each clinical question. The Table of Contents can be used to navigate among the different evidence summaries. The Annex contains supplementary information to support the evidence, including forest plots and Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) profiles.

The content of this CPG is intellectual property of the Philippine Society of Hypertension (PSH). Kindly cite this guideline appropriately when using any of its contents in your own documents, books, or presentations. A soft copy of this material can be accessed online via the PSH and DOH website. Any queries or concerns regarding this guideline may be directed to PSH by email at phihpn@yahoo.com.

ACKNOWLEDGMENT

This CPG was prepared by the Philippine Society of Hypertension (PSH), Philippine Heart Association (PHA) , Philippine College of Emergency Medicine (PCEM) and Philippine Academy of Family Physicians (PAFP). This guideline development was funded by the Philippine Society of Hypertension and the Philippine Heart Association. The DOH neither imposed any condition nor exerted any influence on the operations and the final output formulation.

The guideline developers undertook extensive technical work in (1) performing a literature search, reviewing, and synthesizing the evidence while ensuring objectivity in each stage of the process, (2) presenting the evidence in the panel discussion, and (3) documenting and writing the final report. They also spearheaded the coordination among various individuals, groups, and committees, and facilitated the *en banc* meetings.

The CPG Steering Committee (SC) was responsible for overall direction, organization, and management and is accountable for the quality of the CPG.

Lastly, we highly value the contribution and participation of our panelists from different sectors of the healthcare system. Our panelists dedicated their time and effort to critically analyzing the evidence reviews and ensuring that generated recommendations align with the sociocultural realities of the local setting.

EXECUTIVE SUMMARY

This Clinical Practice Guideline on the Diagnosis and Management of Severe Blood Pressure Elevation is an output from the joint undertaking of the Philippine Society of Hypertension, Philippine Heart Association, Philippine College of Emergency Medicine (PCEM) and Philippine Academy of Family Physicians (PAFP).

The CPG provides eleven (11) recommendations and four (4) best practice statements addressing key clinical questions on the diagnosis and management of severe BP elevation.

Recommendations are based on the appraisal of the best available evidence on each of the eleven identified clinical questions. The CPG is intended to be used by general practitioners, family physicians, allied health professionals, emergency medical personnel, medical specialists, and any healthcare workers who may encounter adult patients with hypertension whether in the inpatient or outpatient setting.

The guideline development process adhered to the GRADE approach through the Evidence to Decision or EtD2 framework. It included 1) identification of critical questions and critical outcomes, 2) retrieval of current evidence, 3) appraisal and synthesis of the evidence, 4) formulation of draft recommendations, 5) convening of a multisectoral Consensus Panel (CP) to discuss values, preferences, and socioeconomic impact and finalize the strength of the recommendations, and 6) planning for dissemination, implementation, impact evaluation, and updating of this CPG.

The recommendations in this CPG shall be updated after 3 years. If new and compelling evidence arises that will have an impact on relevance and application, the task force may opt to reconvene and update the said guidelines.

Keywords: hypertensive emergency, acute elevated blood pressure, pharmacologic treatment, consensus guidelines

1. INTRODUCTION

Atherosclerotic cardiovascular disease (ASCVD) is one of the leading causes of death in the world. In the Philippines, ischemic heart disease (IHD) remains the most common cause of death while cerebrovascular disease or stroke is the third most common [1]. ASCVD, including IHD and stroke, arise as end-organ complications of prevalent and preventable non-communicable diseases (NCD) and their risk factors. The most common of these NCDs – hypertension, remains a prime modifiable risk factor in the progression of ASCVD and its treatment provides an important opportunity for primary and secondary CV prevention. Optimal management of hypertension has been unequivocally proven to reduce the morbidity and mortality associated with ASCVD [2].

The most recent survey on hypertension in the Philippines, PRESYON-4, showed an increasing trend of prevalence from 22% in the 1990s to 37% in 2021[3]. Of these, only 52% were aware of their diagnosis. While rates of treatment and adherence were found to be 68% and 86%, respectively, the rate of BP control was low at 37%. Furthermore, there remained a high degree of unawareness regarding hypertension, its role in CV morbidity and mortality, and how it can be optimally managed. These findings confirmed that hypertension remains a priority problem for Filipinos and that its control is a low-hanging fruit that will have far-reaching beneficial effects for Filipinos in decades to come.

Given the J-shaped correlation of hypertension with ASCVD [4-6], much attention has been paid in particular to the management of acute and severe elevations in BP among patients with and without known hypertension. The 2017 ACC/AHA guideline for hypertension among adults defined systolic blood pressure (BP) >180 mm Hg or diastolic BP >110-120 mm Hg, accompanied by new or worsening hypertension-mediated organ damage (HMOD) as a “hypertensive emergency” [7,8]. The most recent ESC guidelines define hypertensive emergency similarly as BP of ≥180/110 mmHg associated with acute HMOD [9]. Similar elevations in BP but without HMOD have been labeled “hypertensive urgency” by various authors and mentioned in several guidelines [10.] Both hypertensive urgency and hypertensive emergency are grouped under the unofficial umbrella term “hypertensive crisis,” a concept which has recently evolved to a more inclusive term “acute severe BP elevation.”

Hypertensive emergency in particular has been linked with higher rates of admission and mortality in observational studies [11]. However, evidence on the management of hypertensive emergency is mixed [12,13]; it cannot be ascertained if the benefit observed is mostly attributable to the treatment of the severe BP elevation or management of the underlying condition such as ACS or stroke, which often includes lowering BP as well [14,15]. In contrast, there is no evidence that treating hypertensive urgency in the emergency setting offers any distinct benefit in reducing MACE compared to routine outpatient management of chronic hypertension, hence, additional treatment beyond outpatient medication introduction or modification and tailored follow-up is not recommended by the AHA or ESC.

Due to the lack of evidence of relevance, the terms “hypertensive crisis” and “hypertensive urgency” have been endorsed for retirement [15,16]. Furthermore, the AHA 2024 Scientific Statement on the Management of Hypertension in the Acute Care Setting, recommends the transition of the terms “hypertensive crisis” and “hypertensive urgency” to “markedly elevated BP” and “asymptomatic markedly elevated BP”, respectively. In alignment with this terminology shift, we will be referring to the concept previously known as “hypertensive urgency” as “acute severe hypertension.” For the concept described by the term “hypertensive crisis,” we will use the term “severe BP elevation” since the conditions comprising it are sufficiently different and warrant varying approaches to management. The term “hypertensive emergency” is retained throughout the guideline as evidence supports emergent treatment for this group of conditions.

Despite this important shift in terminology, we acknowledge that most literature on these concepts still use the terms “hypertensive urgency” and “hypertensive crisis,” hence we retained their use only for the search strategies of our literature reviews. Medical literature has yielded a mix of evidence on the optimal approach to managing patients with severely elevated BP; hence, the Task Force created this guideline to evaluate the best available evidence and inform healthcare providers on the critical aspects of the diagnosis and management of these conditions.

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2. GUIDELINE DEVELOPMENT METHODOLOGY

2.1 Organization of the Process

Following the international standards and as outlined in its Manual for CPG Development, the DOH divided the guideline development process into four phases: 1) preparation and prioritization, 2) CPG generation, 3) evidence appraisal, and 4) implementation [1].

In the preparation and prioritization phase, the Steering Committee (SC) set the CPG objectives, scope, target audience, and clinical questions. The SC identified the evidence review experts (EREs) through their contacts in recognized hospitals, medical schools, and professional societies and groups. The EREs were tasked with performing the literature review and appraisal and drafting the initial recommendations for each clinical question. The evidence summaries were then presented to the Consensus Panel (CP) members to finalize the recommendations.

The CP was comprised of representatives from different sectors of the healthcare system. They were tasked to review the evidence summaries and to revise and finalize the recommendations during *the en banc* meetings. In the meetings, CP members discussed the sociocultural and economic considerations surrounding the recommendations, and deliberated amongst themselves to produce a consensus on each recommendation with corresponding certainty of evidence and strength. When quorum could not be reached during *en banc* meetings, the modified Delphi approach was used to reach a consensus.

2.2 Creation of the Evidence Summaries

The clinical questions were developed using the PICO (population, intervention, comparator and outcome) format. The EREs performed literature searches and appraised international and regional practice guidelines related to hypertensive crises, including but not limited to those of the European Society of Cardiology, American Heart Association, American College of Cardiology, Asia-Pacific Society of Hypertension, and the International Society of Hypertension among others. If the relevant CPGs were of good quality, released within 5 years, and were performed according to the GRADE approach, the evidence summaries of those CPG were adopted.

If the search for existing CPGs was unsuccessful, a de novo systematic literature search for relevant studies was done involving electronic databases including but not limited to MEDLINE via PubMed, CENTRAL, and EBSCO. Relevant local databases and websites of medical societies, as well as trial registers were also included in the search. Keywords used in the search were based on the PICO (MeSH and free text) of each clinical question (please see appendix). The EREs also contacted authors of related articles to verify details and identify other studies for appraisal, if needed.

At least two EREs worked on each clinical question and each ERE performed selection of articles individually using PICO as inclusion criteria. A third expert was available to facilitate a consensus when the two EREs disagreed on study inclusion. Once included studies were finalized for each clinical question, the EREs independently appraised the directness, methodological validity, results, and applicability of each article. RevMan, STATA, and GRADEPro were used for the evidence synthesis of important clinical outcomes for each question. The EREs generated evidence summaries for each of the eleven (11) clinical questions (Table 1).

Table 1. Clinical questions Generated by the Steering Committee

1. Among hypertensive patients, what are the focused clinical history and physical exam (PE) findings to consider the patient as acute severe hypertension?
2. Among patients with hypertensive urgency, what is the effect of antihypertensive drugs in reducing blood pressure and other cardiovascular clinical outcomes (mortality, stroke, and MI)?
3. Among patients with hypertensive urgency, what drug to use in reducing blood pressure (Sublingual/Oral – class of drugs)?
4. Among patients with Hypertensive Urgency, what is the effect of gradual lowering (24hrs) of blood pressure in reducing cardiovascular mortality, stroke, and MI?
5. Among patients with Hypertensive emergency, what is the effect of rapid lowering (1hr) of blood pressure in reducing cardiovascular death, Stroke and MI?
6. Among patients with hypertensive emergency, what is the effect of IV Nitrates in reducing blood pressure and other cardiovascular clinical outcomes (mortality, stroke, and MI)?
7. Among patients with hypertensive emergency, what is the effect of IV Nicardipine in reducing blood pressure and other cardiovascular clinical outcomes (mortality, stroke, and MI)?
8. Among patients with hypertensive emergency, what is the effect of IV Labetalol in reducing blood pressure and other cardiovascular clinical outcomes (mortality, stroke, and MI)?
9. Among patients with hypertensive emergency what is the effect of IV Hydralazine in reducing blood pressure and other cardiovascular clinical outcomes (mortality, stroke, and MI)?
10. Among patients with hypertensive urgency, what are the indications for referral to a hospital facility?
11. Among patients with hypertensive emergency, what drugs can be given in the primary care setting while the patient is for transfer to a hospital?

Each evidence summary included evidence on the burden of the problem, effect on clinical outcome of interest or diagnostic performance for a condition of interest, benefits, harm, and socioeconomic impact of the intervention or diagnostic procedure. The Certainty of Evidence for each literature review was assessed using the GRADE approach [2].

2.3 Scope and Purpose of the Guidelines

The task force on the "Philippine Clinical Practice Guidelines on the Diagnosis and Management on the Acute Severe Elevation of Blood Pressure" considers this landmark effort as the very first attempt to capture key practice recommendations on hypertensive crisis in a single guide document. The guideline also takes into consideration existing international recommendations on diagnosis and treatment but more importantly appraises and applies such information using the lens of a Filipino healthcare provider. The clinical questions which were initially created by the steering committee were presented and updated during the consensus panel. This is particularly in the change of terminology in using hypertensive urgency to acute elevated blood pressure.

2.4 Target Population

The target population of the guidelines are individuals who present at the emergency department with acute elevation of blood pressure with or without co-morbidities, who needs immediate management. The patients may or may not have hypertension mediated organ damage which will differentiate them from a hypertensive emergency or not. Management will be different for each patient. The task force is cognizant of the more robust scientific evidence for therapies directed towards this classification of hypertensive crisis.

2.5 Intended Users

The expected main end-users of this Clinical Practice Guidelines are Filipino healthcare professionals who diagnose and manage Filipino adults with acute elevation of blood pressure. This guideline will make an impact on the implementation of Universal Health Care. The main intended users of this guideline are the primary care physicians, general practitioners, emergency medicine specialists, family medicine physicians and the internists who will serve as frontliners in the care of the Filipino patients.

Specialists (e.g. cardiologists, nephrologists, endocrinologists) and healthcare workers in the allied professions (e.g. nurses) who are involved in the care of patients acute elevated blood pressure may also benefit from the recommendations stated in this practice guideline. Finally, administrators, policy-makers and other non-medical stakeholders (PhilHealth, HMOs, etc.) may use this document as reference for the development of their respective administrative agenda or reimbursement policies, as the document captures local information and insight regarding burden of disease and socioeconomic impact of various testing modalities and treatment options.

2.6 Composition of the CPG

The CPG Task force is composed of the steering committee, the technical working group, which includes the technical coordinators, the evidence review experts and technical facilitator, the consensus panel, and the conflict of interest review committee. The members of the core group include cardiologists, hypertension specialists, internal medicine specialists, clinical epidemiologists, family medicine practitioners, emergency medicine specialists, nephrologists, neurologists, and endocrinologists who are experts in the field of hypertension.

The steering committee (SC) is the main group that spearheaded the development of this CPG. The SC held several regular meetings to discuss organizational, budget, methodological and scientific content issues. It identified the individuals who were to be part of the other committees. They coordinated evidence summaries with the EREs, facilitated ERE and CP meetings, oversaw the writing, editing and finalization of the CPG, forwarded the final CPG manuscript for external review, and finally, submitted the CPG to the DOH for approval.

The Technical Coordinators (TCs) identified, reviewed, and summarized existing CPGs, and evaluated them for possible adaptation. They also coordinated the activities of the EREs, provided guidance to the EREs regarding evidence synthesis and recommendation formulation, and assisted in the writing and subsequent review of the manuscript. The TCs also coordinated with the technical facilitator in clarifying the clinical issues addressed by the research questions and prioritized for consensus panel discussion and decision-making.

The evidence review experts (EREs) were tasked with reviewing existing CPGs, creating evidence summaries, and drafting evidence-based recommendations. They developed the PICO questions in coordination with the Lead CPG developers.

Prior to the creation of the guidelines, a formal meeting and consultation was conducted to determine the priority topics and questions to be included in the CPG. The technical coordinators and the SC leads presented the initial clinical questions, and the views of the consensus panel were sought before coming up with the initial clinical statements. A patient who suffered from hypertensive emergency, and a clinic nurse were included in the meeting.

The SC, TCs, EREs and the multisectoral consensus panel (CP) agreed on PICO questions to be submitted for approval by the National Practice Guideline committee. The panel also prioritized the critical and important outcome measures prior to finalizing the PICO questions, such that stakeholders' values and preferences were incorporated. Furthermore, the panel reviewed evidence summaries and draft recommendations, and voted as well on the recommendations of the CPG. Discussion and consensus-building during the *en banc* consensus panel meetings were moderated and guided by a technical facilitator.

The conflict of interest review committee reviewed the possible conflicts of interest of everyone involved in the development of this guideline.

After identifying the members of all the committees, the steering committee convened the consensus panel (CP) and considered the possible conflicts of interest of each panel member. To ensure fairness and transparency, the selection process for the composition of the CP was guided by the DOH manual.[1] Content experts and other key stakeholders were invited to join the CP. The key stakeholders included policy makers, patient advocates, allied medical practitioners, and physicians from different settings (e.g., public primary care, private practice, occupational health settings). The physicians were members of different medical societies, namely: Philippine College of Physicians, Philippine Heart Association, Philippine Medical Association, Philippine College of Emergency Medicine, and Philippine Academy of Family Physicians. A nurse and a hypertensive patient were invited as voting members of the CP. The task force ensured that they were adequately guided prior to CP sessions so that any technical jargon or issues would be well-comprehended, and that relevant sentiments were appropriately voiced out during CP sessions. An open line between the patient and the task force, through its technical coordinators and technical facilitator, was maintained, and this greatly assisted the patient as well in appreciating the discussions that ensued. No clear distinction was made by the task force regarding patient groups, and the SC selected a patient who is hypertensive, on maintenance medications and have been hospitalized for an emergency elevation of blood pressure.

A representative of the DOH was also invited to sit in the *en banc* meetings as an observer during the proceedings and would be asked to clarify if there are queries from the CP members regarding policies of the department.

2.7 Management of Conflicts of Interest

All members of the CPG Task Force such as the Steering Committee (SC), Evidence Review Experts (ERE), Technical Coordinators (TC), Consensus Panelists (CP), Technical Facilitator, Technical Writer, and Administrative Officer underwent screening for conflicts of interest through their submitted curriculum vitae and accomplished the prescribed declaration of conflict of interest forms. The COI Review Committee of the CPG Task Force was composed of three individuals recommended by the Steering Committee and approved by the National Practice Guideline (NPG) COI Review committee. The Steering Committee and the Task Force COI Review Committee (TFCOIRC) underwent training on screening and managing conflicts of interest. The TFCOIRC reviewed the CVs and COI forms of all the members of the task force and made recommendations regarding each CPG Task Force member’s extent of participation.

The Steering Committee (SC) invited relevant professional organizations to nominate individuals who can become part of the consensus panel. Selected experts, by virtue of their training background and experience in critical appraisal and guideline development, were also invited by the SC to serve as EREs. Such nominees were initially screened by the SC for any major and obvious conflicts of interest that may introduce bias in their decisions. Those who passed the initial adjudication underwent screening by the Task Force COI Review Committee. Certain consensus panelists were prohibited from voting on certain clinical questions if they possessed significant potential COIs relevant to the said questions. The process of COI evaluation was guided by the COI algorithm (Fig. 2) crafted by the UP-NIH National Practice Guideline Project Team. Each task force member's curriculum vitae and declared conflicts of interest were assessed by the TFCOIRC and a decision on his/her participation was made, as follows: Approved with no constraint (Status A); Broadcast conflict at discussion (Status B); Cannot vote regarding certain guideline question/s (Status C); Disqualified or Disqualified for specific guideline question/s (Status D).

For the first batch of SCs and CPs, major COIs were identified in a significant number of nominees. These recommendations were submitted to the Steering Committee which, in turn, informed the members of the task force regarding the outcome of the COI assessment and the recommendations for the appropriate management of identified potential conflicts of interest. To manage this, the SC recruited additional members without major COI to ensure that the majority of the members of the SC and the CP had no conflicts of interest. Declaration of COI assessment was done at the start of every meeting.

Each nominee was required to fill out and sign a declaration of interest form and submit their curriculum vitae. The COI committee screened the nominees for any potential COI that may bias their decisions for this CPG. Those with significant potential COI were not allowed to join the roster of CP members. The COI assessment results are available per request with the Philippine Society of Hypertension by email at phihpn@yahoo.com.

2.8 Evidence Synthesis

A systematic search of electronic databases such as MEDLINE via PubMed, CENTRAL, and Google Scholar for relevant literature was performed. Guidelines on acute elevated blood pressure management were identified and underwent assessment using the AGREE II tool.[2] The following guidelines were selected based on the search terms of chronic heart failure guidelines and were approved by the steering committee. The following relevant guidelines were included:

- 2020 AHA/ACC Guidelines in the Management of Hypertension
- 2020 Philippine Practice Guidelines in Hypertension
- 2024 ESC Guidelines on the Management of Elevated Blood Pressure
- 2020 International Society of Hypertension
- 2017 British and Irish Hypertension Society Position Paper on the Management of Hypertensive Crisis

De novo systematic reviews and meta-analysis were done for each question as the guidelines were not able to answer the clinical questions developed by the SC. Databases such as MEDLINE via Pubmed, Google Scholar, and HERDIN were utilized to search the relevant literature. Meta-analysis and systematic review sites such as COCHRANE reviews and PROSPERO were used. Local databases were also reviewed for studies relevant to the clinical questions.

Keywords were based on PICO (population, intervention/exposure, comparison, outcomes), MeSH and free text, set for each question. The ERE contacted authors of related articles to verify details and identify other research studies for appraisal. The specific search terms used by the Evidence Reviewers to search for the relevant literature are found in each annex for the respective topics.

The studies included were prospective, cohort, cross-sectional, and systematic reviews for questions on diagnosis. For questions on therapy, meta-analyses, randomized controlled trials (RCTs), and network meta-analyses were included.

A data extraction tool used by Raftery et al (2015) was utilized by the evidence reviewers. The table includes the type of design, description of the clinical trial, type of RCT (superiority, inferiority, equivalence), results of the study, and other relevant information seen in the tool. Two experts filled out the extraction tool while a third reviewer was called to resolve discrepancies. Similarly, the data extraction tool was also used for clinical practice guidelines.

Health economic outcome research (HEOR) and cost-effectiveness studies, both international and local, were included in the search to answer the cost portion of the evidence-to-decision framework. If there were no studies on cost-effectiveness, the evidence reviewer checked for the cost of the diagnostic test in hospitals or diagnostic centers, and the cost of the drug in the national formulary or in drugstores. Studies on health equity and patient preferences and values were also included in the search terms to answer the EtD table. Studies on patients' views and preferences regarding interventions were also searched and evaluated.

2.9 Study Quality Assessment

Appraisal tools for clinical studies using the Newcastle Ottawa Scale[11] or QUADAS[12] were utilized by the EREs in developing *de novo* assessments. Systematic reviews and meta-analyses using the Cochrane approach were used. Relevant local databases and websites of medical societies were also utilized in the search.

2.10 Data Synthesis

Evidence reviewers appraised the directness, methodological validity, results, and applicability of each relevant clinical study or guidelines included. RevMan, STATA, and GRADE Pro were used for the quantitative synthesis of important clinical outcomes for each question. The ERE-generated evidence created summaries for each of the fourteen questions. Each evidence summary included evidence on the burden of the problem as well as on diagnostic performance, benefits, harm, and the social and economic impact of the screening test/intervention. Evidence/information that will facilitate decision-making (i.e., cost of screening test, cost-effectiveness studies, qualitative studies) were also included in the evidence summaries. The important step in the creation of this clinical practice guideline (CPG) was determining the strengths of the recommendations. The Grading of Recommendations Assessment, Development, and Evaluation (GRADE) approach was used in assessing the recommendations. (Table 2)

Table 2. Basis for Assessing the Quality of the Evidence using GRADE Approach

Certainty of Evidence	Interpretation
High	We are very confident that the true effect lies close to the estimate of the effect.
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect but there is a possibility that it is substantially different.
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.
Very Low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect.
Factors that lower quality of the evidence are: ● Risk of bias ● Important inconsistency of results ● Some uncertainty about directness ● High probability of reporting bias ● Sparse data/Imprecision ● Publication bias Additional factors that may increase quality are: ● All plausible residual confounding, if present, would reduce the observed effect ● Evidence of a dose-response gradient ● Large effect	

The SC, EREs and CP identified the outcomes that were used in deciding the recommendations for the clinical questions as critical, important, or of low importance; only critical and important outcomes were selected.[1] After identification, the CP validated and voted on the rating of the outcomes. All the panelists, including the patient advocate and the cardiac rehabilitation nurse, prioritized and rated the clinical outcomes that were either critical or important (Table 3).

Table 3. Cardiovascular Outcomes GRADE Score

OUTCOME	SCORE PRIORITY	RANK
Cardiovascular Mortality	9	Critical
All-cause Mortality	9	Critical
Worsening Heart Failure	9	Critical
Hospitalization from elevated blood pressure	8	Critical
Symptom reduction	8	Critical
Quality of Life	8	Critical
Hypotension	8	Critical
Arrhythmia	8	Critical
Deteriorating Renal Function	7	Important

2.11 Formulating Recommendations

The evidence was assessed by the reviewers using the GRADE approach. The recommendations were based on the quality of evidence, trade-off between the benefits and harm of the test or drug, cost-effectiveness, applicability, availability, feasibility, equity, resources and uncertainty. Factors that lowered the quality of evidence, such as publication bias, inconsistency of results and reporting bias, were considered. An Evidence to Decision (EtD) framework was used to process the gathered evidence and come up with decisions and recommendations. This was created by the GRADE working group to make clinical recommendations, coverage decisions, and public health recommendations. The framework was built to assess the strength of recommendations.[13]

The criteria used in the EtD framework that was followed by this guideline include priority of the problem, test accuracy, benefits and harms, certainty of evidence, outcome importance, balance, resource use, equity, acceptability, and feasibility. These were used by the evidence reviewers to determine the strength of recommendation and by the consensus panel in weighing the evidence and the possibility of recommending it.

The strength of each recommendation (i.e. strong or weak) was determined by the panel by considering all the factors mentioned above. Strong recommendation means that the panel is “confident that the desirable effects of adherence to a recommendation outweigh the undesirable effects” while weak recommendation means that the “desirable effects of adherence to a recommendation probably outweigh the undesirable effect” but the panel cannot say so with confidence. A weak recommendation means that the physician may either follow the recommendation or use another option of diagnostic test or treatment management.

The recommendation for each question and the strength of each recommendation were determined through voting. A consensus decision was reached if 75% of all CP members agreed. In case that consensus was not reached in the first voting, questions and discussion were encouraged. Two further rounds of voting on an issue were conducted. Evidence-based recommendations were based on input arrived at by consensus in the *en banc* discussions. The recommendations presented by the evidence-based reviewers were moved forward and revised depending on the decision of the consensus panel during voting. If the decision has not been reached even after three rounds of voting, a modified Delphi approach was conducted to be spearheaded by the lead of the steering committee.

There were recommendations that have been graded LOW based on the evidence presented by the reviewer; however, during the *en banc* meeting, the consensus panelists felt that these recommendations needed to be upgraded to a STRONG recommendation, based on factors such as feasibility, equity, and applicability in the local setting. Thus, the CPG included recommendations that have low level of evidence but with a strong recommendation.

In cases where direct evidence was lacking and the consensus panel deemed it necessary that a statement be articulated regarding the intervention in question, a good practice statement was included. A good practice statement implies that there is general agreement that the intervention will do more good than harm.

Table 4. GRADE certainty of evidence definitions and factors that reduce or increase the quality of evidence

Grade	Definition
High	We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different
Low	Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect.
Very Low	We have very little confidence in the effect estimate: The true effect is likely to be substantially different from the estimate of effect

Factors that can reduce the quality of the evidence	
Factor	Consequence
Limitations in study design or execution (risk of bias)	↓ 1 or 2 levels
Inconsistency of results	↓ 1 or 2 levels
Indirectness of evidence	↓ 1 or 2 levels
Imprecision	↓ 1 or 2 levels
Publication bias	↓ 1 or 2 levels

Factors that can increase the quality of the evidence	
Factor	Consequence
Large magnitude of effect	↑ 1 or 2 levels
All plausible confounding would reduce the demonstrated effect or increase the effect if no effect was observed	↑ 1 level
Dose-response gradient	↑ 1 level

2.12 Plan for Dissemination and Implementation

The SC discussed with relevant stakeholders such as the DOH and medical societies the preparation of a dissemination plan to actively promote the adoption of this guideline with strategies for copyrights.

The dissemination and implementation plan of this CPG includes the following:

1) writing of an Executive Summary that will be submitted for publication in a reputable local peer-reviewed medical and cardiology journals. The summary will include a quick reference guide of all the recommendations and the algorithm to be used by primary care physicians, cardiologists and emergency medicine specialists, and allied medical professionals who are managing acute elevation of blood pressure.

2) pilot-testing of the recommended guideline-based pathway of care in identified hospitals in select key cities after the publication of the executive summary. Educational webinars on the CPG for primary care physicians will be held. A pre-test/post-test evaluation will be conducted after the medical education. The pilot-testing will run for six months, and a survey will be conducted to measure success of the implementation. A focus group discussion will also be conducted among the primary care physicians to assess their improvement in the management of acute elevation of blood pressure. Then, with the approval and endorsement of the DOH, the guidelines will be implemented nationwide.
3) Stakeholder medical societies (PHA, PCP, PAFP, PMA, PCEM) will be asked to participate in the dissemination of the guidelines during their annual scientific meetings and in their websites. Once the article is published, the CPG will be presented during scientific conventions and be made available on the websites of the DOH, PHC, different medical societies and organizations. Other avenues of dissemination will include press conferences, social media sites, and professional society conventions.
4) Electronic clinical decision support tools will be designed and developed to assist primary care physicians in the implementation of the guideline recommendations.
5) Rapid assessment method of evaluating adherence to guideline recommendations may be done after one year of dissemination efforts in selected and representative outpatient practice groups by reviewing outpatient records. Through email.

A full copy of this document will be sent to the Department of Health for appraisal and publication on the DOH website once deemed a DOH-approved CPG. The Disease Prevention and Control Bureau will transmit copies of the CPG to relevant stakeholders.

The following criteria will be included to assess the success of implementation of the CPG for acute elevation of blood pressure. These were adapted from the guideline checklist developed by National Health Guidelines of Australia[14]. This checklist includes the ff: 1) systems are in place to support comprehensive care for heart failure patients, 2) awareness and distribution of the guidelines among the healthcare professionals, 3) clinical education on the guidelines, and 4) clinical care is aligned with the guidelines.

2.13 Applicability of the Guidelines

2.13.1 Facilitators and Barriers

Prior to the start of guideline development, the steering committee members discussed possible facilitators and barriers to the diagnosis and management of acute elevated blood pressure in the process of deciding on the coverage of the contemplated clinical practice guideline. EREs and CP members were also engaged in such discussion during the organizational meeting of the Task Force.

These guidelines address the needs of primary care physicians and emergency medicine specialists who are first to see patients suffering from acute elevated BP in the outpatient setting. Successful implementation of the guidelines include active dissemination of the knowledge on hypertensive emergency management and measuring its impact to clinical practice among all stakeholders. Specialty societies, such as the Philippine Heart Association and Philippine Society of Hypertension, and cardiology training institutions will be tapped to educate and train primary care physicians. The task force proposes that hypertensive emergency medications recommended in these guidelines be approved for inclusion in the national formulary to improve utilization of the guidelines. However, there may be barriers in the management of acute blood pressure elevation. There is limited availability of specialty centers and support centers for heart failure, such as heart failure clinics and cardiac rehabilitation centers. This CPG recommends that care for patients with acute elevated blood pressure be coordinated with cardiologists in their hospitals. There may also be a need for hypertension clinics in the country handled by specialists who will manage difficult to treat hypertensive crisis.

2.13.2 Guideline Monitoring and Evaluation

The National Institute of Health, in cooperation with the steering committee and the task force, will create guidelines on monitoring of the guidelines, from publication, dissemination, and application in clinical practice. The task force hopes that such guideline recommendations will be properly cascaded to intended users through close collaboration between the national government, key professional societies, and local health administrators. Strategies may be developed to determine real-world use of such guidelines, such as quality of care studies and validation of implementations studies, looking into adherence to key recommendations and impact of their application, whether through scorecard metrics or coordinated research, such as rapid assessment or quality of care studies. Such evaluations may be planned based on pre-specified intervals, depending on the current need of the health industry.

2.14 External Review

The Steering Committee invited 3 external reviewers to comprise the Task Force Guideline External Review Panel. The external reviewer panel was composed of an endocrinologist/clinical epidemiologist, a clinical cardiologist, and a clinical epidemiologist/health economist, who reviewed the initial manuscript of the guidelines. All three external reviewers used 2 tools – 1) the AGREE-II checklist, to evaluate the methodological aspects of CPG development, including its reporting, and 2) the AGREE-REX checklist, to assess the technical aspects and implementability of the CPG recommendations.[15] The SC held a meeting after receiving the external reviews and recommendations which were reviewed and considered in the revision of the final draft of the manuscript before submission to the central committee. Below is a summary of the reports of each of the three external reviewers:

External Reviewer 1

Limited discussion on how patient values/preferences were considered during guideline development. More engagement with patient advocacy groups would be beneficial, although a patient advocate was included in the consensus panel. Some discussion is provided, but challenges in implementation, especially in resource-limited settings, need further elaboration.

External Reviewer 2

The guideline developers tried to incorporate economic analysis in every recommendation. However, there is a need to be explicit on the following: 1) perspective of the analysis and 2) cost being synonymous with charges or price. Moreover, in trying to follow the template or format of the manuscript, there were several times that the developers filled up the content of the “Cost” section with “paucity of cost-effectiveness studies or analysis (CEA), etc.” which gives the impression that every recommendation/statement needs to have a “cost-effectiveness study”.

External Reviewer 3

The guideline can be used in our local setting even if there are barriers or limitations identified. But having these barriers and limitations identified, the relevant societies, stakeholders and government institutions can work in another venue/s to help “facilitate” the application and adoption of this guideline and the development of researches that are pertinent to implementation of the guideline (e.g. cost effectiveness studies that Phiilhealth can use)

2.15 Cost Implications of the Guidelines

Evidence review experts were asked to include in their search strategy cost-effectiveness studies on the diagnostic tests and medical treatments for their evidence review of the statement recommendations. The consensus panelists also considered the cost implications of each recommendation prior to voting. During some of these discussions where the ERE was able to present cost-effectiveness studies, the steering committee deemed it to be challenging to present to the Consensus Panel, given the time restrictions. Furthermore, some of the EREs were not able to retrieve such studies and are not capable of performing de novo cost-effectiveness studies. One of the external reviewers is a clinical epidemiologist and a health economist who also checked on the method of reviewing cost-effectiveness studies and offered suggestions. The Steering Committee agreed that this knowledge gap may be addressed in subsequent updates of the CPG.

We are anticipating that dissemination and implementation of the guidelines will require costs. The steering committee will come up with a budget on the implementation, dissemination, assessment and evaluation of the CPG. This will be discussed with the central committee, the National Institutes of Health and the Department of Health.

Some of the medications that are recommended are not included in the national formulary and maintenance treatment for hypertensive emergencies at current retail prices may not be affordable to most patients. The guideline developers will hold dialogues with the Health Technology Assessment Committee who are in charge with recommending medications to be part of the national formulary to include the recommended drugs and diagnostic tests so the government can procure these medications and make them accessible to hypertensive patients.

2.16 Editorial Independence

The funding for the development of this CPG was provided by the Philippine Heart Association, Philippine Society of Hypertension, Philippine College of Emergency Medicine, and Philippine Academy of Family Physicians in preparation for the implementation of Universal Health Care. These societies did not influence the guideline development. Clinical practice guideline recommendations were solely and independently formulated by the Hypertensive Crisis Task Force.

2.17 Guideline Update

A proposed updating process framework for CPGs starts with assembling a group responsible for updating the CPG, consisting of methodologists and experts, similar to the original composition of the CPG task force.[16] New issues and technologies may require experts from new fields. The actual updating process starts with a systematic search for new relevant evidence. The DOH recommends that updates may be needed if there are identified gaps in the current knowledge on the diagnosis and management of hypertensive emergencies, newly released evidence from large scale studies or clinical trials, approval of new interventions or therapies, changes in critical or important outcomes and values placed on outcomes, or changes in resources available for health care.[1] A fixed time frame from two to five years had been reported in previous reviews of methodological handbooks.[17] After identifying new relevant evidence, the effect or impact of this new evidence should next be assessed to determine the need for an update. This assessment should include whether this new evidence alters the validity of the current recommendations. When the need for an update is agreed upon, the new evidence is incorporated and the current recommendations are revised accordingly. A multidisciplinary external review by a panel with members not involved in the development or updating of the CPG, and including clinical and methodological experts, should also be conducted. Finally, when approved by the DOH, a summary of the updated recommendations or *de novo* recommendations should be published.[16]

Thus, updates will be planned three years after publication. The steering committee will organize a CPG update group that will meet annually to assess latest clinical trials and observational studies that may impact the practice of heart failure management in the country. As implementing agency, the Philippine Society of Hypertension will coordinate with the SC in the conduct of this annual review and will contact the National Institutes of Health and Department of Health for possible collaboration in updating the guidelines. Subsequently, a technical working group to look into the latest evidence, and a consensus panel to deliberate on the new recommendations, with members who may have been part of the original CPG task force, will be convened.

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SUMMARY OF RECOMMENDATIONS

Recommendation/Best Practice Statement	Certainty of Evidence	Strength of Recommendation
Among patients with SBP >= 180 mmHg or DBP <=110 mmHg without signs and symptoms of hypertension-mediated organ damage, we recommend the term **acute severe hypertension** instead of hypertensive urgency.	Best Practice Statement	
Among adult patients presenting with an elevated blood pressure of SBP >=180 mmHg or DBP >=110 mmHg, we recommend to elicit the presence of at least one of the main symptoms: • focal neurologic complaints • visual impairment • dyspnea • chest pain • headache when deciding to evaluate further for the diagnosis of hypertensive emergency.	Very low	Strong
Among patients with severe BP elevation (SBP >=180 mmHg or DBP >=110 mmHg), we suggest screening for signs and symptoms of hypertension-mediated organ damage to rule out hypertensive emergency, which will warrant referral to a hospital. Older age (>60 years old), male sex, presence of chronic kidney disease, and proteinuria are risk factors among patients with acute severe hypertension, which necessitate close monitoring but can be managed in the outpatient setting.	Low	Weak
Among adult patients with acute severe hypertension whose BP does not respond after two hours of rest, we suggest the use of oral antihypertensive medications* to decrease BP. *initiation or intensification of existing oral antihypertensive medication	Very low	Weak
Among adult patients with acute severe hypertension, we suggest out-of-office BP monitoring with home BP or 24-hour ambulatory BP monitoring and follow-up within one week.	Best Practice Statement	
	Very low	Strong

Among adult patients with acute severe hypertension, we recommend the use of oral antihypertensive medications over sublingual antihypertensive medications to lower BP.		
Among adult patients with acute severe hypertension, we recommend gradual reduction of BP over 24 to 48 hours over immediate lowering to normal levels to reduce all-cause mortality, MACE, and ED re-visits.	Very low	Strong
Among adult patients with hypertensive emergency and intracerebral hemorrhage (ICH), we suggest rapid BP lowering to SBP<140 mmHg within 1 hour to prevent worsening of stroke.	Low	Weak
Among adult patients with hypertensive emergency and complications other than ICH, we suggest rapid BP control to lower SBP as indicated in the table below. (see text for table)	Best Practice Statement	
Among adult patients with hypertensive emergency and a cardiovascular event**, we recommend the use of IV nitrates to lower BP and reduce all-cause mortality. **acute coronary syndrome, acute cardiogenic pulmonary edema	Low	Strong
Among adult patients with hypertensive emergency, we recommend the use of IV nicardipine to lower BP and reduce hospital length of stay.	Low	Strong
Among adult patients with hypertensive emergency, we suggest not to use IV labetalol to lower BP.	Very low	Weak
Among non-pregnant adult patients with hypertensive emergency, we suggest not to use IV hydralazine to lower BP.	Low	Weak
Among adult patients with hypertensive emergency in the primary care setting while the patient is for transfer, we suggest the use of antihypertensive medications to reduce BP (IV preferred but if not available, oral).	Very low	Weak

Among adults with hypertensive emergency in the primary care setting while the patient is for transfer, we suggest referral to a health facility capable of monitoring and safely lowering BP.	Best Practice Statement

3. RECOMMENDATIONS AND PANEL DISCUSSION

3.1 CLINICAL HISTORY AND PHYSICAL EXAMINATION

BEST PRACTICE STATEMENT 1

Among patients with SBP equal to or greater than 180 or DBP of equal to or greater than 110 without signs and symptoms of hypertension-mediated organ damage, we recommend the term **acute severe hypertension** instead of hypertensive urgency.

RECOMMENDATION 1

Among adult patients presenting with an elevated blood pressure of SBP >=180 mmHg or DBP >=110 mmHg, we recommend to elicit the presence of at least one of the main symptoms:

- focal neurologic complaints
- visual impairment
- dyspnea
- chest pain
- headache

when deciding to evaluate further for the diagnosis of hypertensive emergency.

(certainty of evidence very low, strength of recommendation strong)

CONSIDERATIONS

Prior to discussing the clinical question and draft recommendation, the representative from the PHA surfaced the issue of the evolving terminology for severe hypertension and highlighted the paradigm shift away from the terms “hypertensive crisis” and “hypertensive urgency.” This ongoing shift is due to evidence of lack of benefit for urgent treatment of patients with so-called hypertensive “urgency” and the possible harms of rapid BP lowering among these patients. This led to the drafting of a new best practice statement, which was unanimously accepted by all CP members after one round of voting.

The SC discussed the scope of the clinical question, which initially included only adults with a prior diagnosis of hypertension and presenting at the ED. Upon dialogue, all SC members concurred on the relevance of revising the scope of the clinical question to include **all adults with severely elevated BP with or without a known diagnosis of hypertension in all settings** after considering the following:

- The high prevalence of hypertension unawareness in the Philippines, which was pegged at 48% of identified persons with hypertension in the PRESYON-4 survey
- The representation of both aware and unaware populations in the best available evidence
- The crucial nature of eliciting the above signs and symptoms regardless of a prior diagnosis of hypertension in both primary care and ED settings

The CP decision was unanimous regarding the positive direction of the recommendation, in favor of using the above symptoms to screen for hypertensive emergency. During the first round of voting on the strength of recommendation, there was divergence on the members' decisions, where representatives from the PHA and PSH voted for a strong recommendation and the rest voted for a weak recommendation. The following points were raised.

- Members who voted for a strong recommendation discussed the high specificity of symptoms based on best available evidence, benefits of identifying cases of hypertensive emergency, and the importance of promptly instituting subsequent life-saving management.
- Members who voted for a weak recommendation brought up the very low certainty of evidence and risks of overdiagnosis, such as overuse of medical tests and services and unnecessary referrals.

During the second round of voting, all CP members agreed that the strength of recommendation was strong despite the very low certainty of evidence, due to the high specificity of the identified symptoms and the great importance of identifying and managing life-threatening hypertensive emergency.

KEY FINDINGS

- There were no significant differences in presenting BP at the emergency room among patients with hypertensive emergency or urgency, hence the degree of BP elevation is not a reliable indicator for differentiating the two conditions.
- Presenting symptoms of focal neurologic complaints, dyspnea, visual impairment, chest pain and headache in addition to elevated BP have high specificity and positive predictive value for the diagnosis of hypertensive emergency.

CHARACTERISTICS OF STUDIES

A systematic review and meta-analysis of eight studies (five prospective and three retrospective studies) with a total of 1,970 patients with hypertensive emergency and 4,983 patients with hypertensive urgency investigated the prevalence of both disease entities, predictive values of specific symptoms, degree of BP elevation, and risk factors, as well as relative frequency of acute HMOD.[1] A second systematic review and meta-analysis analyzed 19 studies looking into non-modifiable (age, sex and ethnicity) and modifiable factors (socioeconomic status [SES], adherence to medical therapy, comorbidities, and substance abuse) of adult hypertensive patients and their impact on the likelihood of hypertensive urgency and emergency. [2]

A retrospective cohort of 718 patients presenting with elevated BP at the ED of a hospital in Turin, Italy looked into the diagnostic accuracy of specific presenting symptoms associated the diagnosis of hypertensive emergency. [3] Specifically, the diagnostic performance of an algorithm proposed by Van der Born and colleagues and endorsed by the ESC in 2019 was evaluated. [4] No diagnostic accuracy studies investigated the sensitivity and specificity of specific clinical signs in establishing the diagnosis of hypertensive emergency.

EFFICACY OUTCOMES

Level of BP elevation and symptoms

Upon presentation at the ED, there was no significant difference in SBP values among patients with hypertensive emergency and urgency (MD 1.4 mmHg; 95% CI -0.8 to 3.6), while the DBP values were found to be slightly higher for hypertensive emergency (MD 2.3 mmHg; 95% CI 0.3 to 4.3). [1]

For the presenting symptoms, a study showed that the algorithm endorsed in 2019 by the ESC for use in diagnosing true hypertensive emergencies has a diagnostic accuracy of 64%, sensitivity of 94%, specificity of 60%, negative predictive value (NPV) of 99%, and a positive predictive value (PPV) of 23%. A strong association was noted between the main symptoms therein and hypertensive emergency after logistic regression analysis (OR: 18.315, 95% CI 7.82 to 42.9). [3] Among the symptoms, focal neurological signs had the highest PPV of 51.6%, followed by dyspnea at 25.8%, visual impairment at 20%, chest pain at 11.3%, and headache at 4.4% (Table 3).

Comorbidities and medical history

The presence of comorbidities such as coronary artery disease (OR 1.654, 95% CI 1.232 to 2.222), congestive heart failure, cerebrovascular disease (OR 1.769, 95% CI 1.218 to 2.571), and chronic kidney disease (OR 2.899,95% CI 1.32 to 6.364) was associated with a higher risk of hypertensive crises in general. [2] In the same study, hypertensive emergency was more common in men (OR 1.390, 95% CI 1.2071 to 1.601), older patients (MD 5.282, 95% CI 0.624 to 3.461), and those with diabetes (OR 1.723, 95% CI 1.485 to 2.000) and hyperlipidemia (OR 2.028, 95% CI 1.642 to

2.505). In contrast, non-adherence to antihypertensive medication and unawareness of the hypertension diagnosis did not increase the risk of hypertensive emergency.

Table 5. Diagnostic performance of signs and symptoms for the diagnosis of hypertensive emergency

Pooled Analysis	Studies	Pooled Estimate	95% CI	Interpretation	Certainty of Evidence
Signs and Symptoms					
SBP	2 retrospective, 5 prospective	MD 1.4mmHg	-0.8mmHg to 3.6mmhg	Does not favor diagnosis of hypertensive emergency	Low
DBP	2 retrospective, 5 prospective	MD 2.3mmHg	0.3mmHg to 4.3mmHg	Favor diagnosis of hypertensive emergency	Low
Symptom-based Diagnostic Strategy (ESC 2019, Van den borg and colleagues)					
Sn	1 Cohort	94%	86% to 89%	Absence of main symptoms rules out hypertensive emergency (High sensitivity, low specificity)	Moderate
Sp	1 Cohort	60%	56% to 64%		Moderate
PPV	1 Cohort	23%	21% to 25%		Moderate
NPV	1 Cohort	99%	97% to 100%		Moderate
Symptom-specific (at least one main symptom + elevated blood pressure)					
Chest Pain					
Sn	1 Cohort	16.46%	9.06% to 26.49%	Favors diagnosis of hypertensive emergency	Moderate
Sp	1 Cohort	88.26%	85.51% to 90.66%		Moderate
PPV	1 Cohort	14.77%	9.17% to 22.93%		Moderate
NPV	1 Cohort	89.53%	88.53% to 90.44%		Moderate
Dyspnea					
Sn	1 Cohort	31.65%	21.63% to 43.08%	Favors diagnosis of hypertensive emergency	Moderate
Sp	1 Cohort	88.73%	86.02% to 91.08%		Moderate
PPV	1 Cohort	25.77%	19.02% to 33.90%		Moderate
NPV	1 Cohort	91.31%	90.02% to 92.44%		Moderate

Neurological focal signs					
Sn	1 Cohort	37.97%	27.28% to 49.59%	Favors diagnosis of hypertensive emergency	Moderate
Sp	1 Cohort	95.31%	93.37% to 96.81%		Moderate
PPV	1 Cohort	49.99%	38.96% to 61.03%		Moderate
NPV	1 Cohort	95.26%	91.27% to 93.67%		Moderate
Headache					
Sn	1 Cohort	7.59%	2.84% to 15.80%	Favors diagnosis of hypertensive emergency	Moderate
Sp	1 Cohort	86.85%	83.99% to 89.38%		Moderate
PPV	1 Cohort	6.66%	3.13% to 13.65%		Moderate
NPV	1 Cohort	88.38%	87.64% to 89.08%		Moderate
Visual Impairment					
Sn	1 Cohort	5.06%	1.40% to 12.46%	Favors diagnosis of hypertensive emergency	Moderate
Sp	1 Cohort	97.50%	95.97% to 98.56%		Moderate
PPV	1 Cohort	20%	7.89% to 42.16%		Moderate
NPV	1 Cohort	89.26%	88.74% to 89.75%		Moderate
Medical History (including co-morbidities)					
Chronic Kidney Disease	19 studies	OR: 2.899	1.32 to 6.364	Increases risk of hypertensive emergency	Very Low
Coronary artery disease	19 studies	OR: 1.654	1.232 to 2.222	Increases risk of hypertensive emergency	Very Low
Stroke	19 studies	OR: 1.769	1.218 to 2.571	Increases risk of hypertensive emergency	Very Low
Diabetes	19 studies	OR: 1.723	1.485 to 2.000	Increases risk of hypertensive emergency	Very Low
Dyslipidemia	19 studies	OR; 2.028	1.642 to 2.505	Increases risk of hypertensive emergency	Very Low

Non-adherence to antihypertensive medication	19 studies	OR: 0.939	0.647 to 1.363	No association with hypertensive emergency	Very Low
Hypertension diagnosis unawareness	19 studies	OR: 0.807	0.564 to 1.154	No association with hypertensive emergency	Very Low

MD: mean difference, Sn: sensitivity, Sp: specificity, PPV: positive predictive value; NPV: negative predictive value; OR: odds ratio

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

Currently, no data is available on Filipino adults' values and preferences including equity and acceptability of history and physical examination in diagnosing hypertensive urgencies and emergencies.

RECOMMENDATIONS FROM OTHER GROUPS

Among the ISH, AHA, and ESH clinical guidelines, there are no specific BP cut-offs used to differentiate hypertensive urgency and emergency. [4,5] The ISH suggests the identification of specific data in the medical history and physical examination to identify HMOD, however they did not specify which specific signs and symptoms can reliably differentiate between hypertensive urgency and emergency. [5]

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ACUTE SEVERE HYPERTENSION

(FORMERLY "HYPERTENSIVE URGENCY")

3.2 INDICATIONS FOR REFERRAL

RECOMMENDATION 2

Among patients with acute severe hypertension (>=180/110), we suggest screening for signs and symptoms of hypertension-mediated organ damage to rule out hypertensive emergency, which will warrant referral to a hospital.

Older age (>60 years old), male sex, presence of chronic kidney disease, and proteinuria are risk factors among patients with acute severe hypertension, which necessitate close monitoring but can be managed in the outpatient setting.

(certainty of evidence low, strength of recommendation weak)

CONSIDERATIONS

During the discussion for this recommendation, the following issues were considered:

- Lack of controlled clinical trials investigating the benefit of referring to urgent care for patients with acute severe hypertension
- Nonrandomized studies showing no benefit in referring to a hospital for urgent care among patients with acute severe hypertension
- In response to lack of evidence for urgent treatment, other professional groups have recommended the retirement of the term "hypertensive urgency" as it contradicts with the non-urgent management of this condition
- Convergence of evidence for this clinical question with the evidence regarding signs and symptoms in Recommendation 3.1, highlighting the importance of eliciting symptoms of hypertension-mediated organ damage to guide decision to refer

During the first round of voting, all CP members unanimously voted in favor of screening for signs and symptoms whose presence would suggest hypertensive emergency and would guide the need for referral. In terms of strength of recommendation, all members voted for a weak recommendation due to the lack of controlled trials on this topic and the low certainty of the best available evidence.

KEY FINDINGS

- There are no direct studies that look into the indications for referral to a hospital facility for patients with hypertensive urgency.
- No evidence on short term benefits were seen for treating patients with hypertensive urgency. Since patients are first seen in the outpatient clinic, signs of acute hypertension-mediated organ damage need to be ruled out especially among those who symptomatic.
- There was no difference seen in major adverse cardiovascular event (MACE) between those who were sent home or referred to a hospital for further

management. Factors that are associated with 3-year mortality were older age (>60 years old), male sex, presence of chronic kidney disease and proteinuria.
- Overall certainty of evidence was low due to observational research design of the included studies.

CHARACTERISTICS OF STUDIES

There were no studies that directly answered the clinical question on the indications for referral to a hospital facility for patients with acute severe hypertension.

Although no direct evidence exists, patients in the outpatient setting with acute severe hypertension and with symptoms such as headache, dizziness, chest pain, shortness of breath, vomiting or blurring of vision warrants further evaluation in the hospital to rule out presence of HMOD. The following tests need to be done for assessment of HMOD: fundoscopy, 12-L ECG, serum creatinine, and dipstick urine test.

There were no studies to show that acute severe hypertension is associated with an acute elevation in cardiovascular risk. There were two observational studies that looked into all-cause mortality and major adverse cardiovascular events (MACE) rates among patients with acute severe hypertension (see Table 4). Factors that were associated with a higher incidence of 3-year mortality were older age (>60 years old), male sex, presence of chronic kidney disease, and proteinuria. [1] However, in another observation study, patients who were referred to a hospital emergency department showed no difference in MACE compared to those who were sent home from the clinic after 6 months of follow-up. [2]

Table 6. Summary of findings for predictors of MACE and all-cause mortality among patients with acute severe hypertension

Outcomes	No. of Studies Population	Result	Interpretation	Certainty of Evidence
All-cause mortality [1]	1 Observational, cross-sectional study at Hanyang University Guri Hospital, Guri-si, Gyeonggi-do, Republic of Korea (N=4,488) Median follow-up 3.1 years	Predictors for 3-year mortality: • Age ≥ 60 years old HR 16.6, 95%CI 6.2 – 44.8, p<0.001 • Male sex HR 1.54, 95% CI 1.22 – 1.94, p<0.001 • History of chronic kidney disease HR 2.18, 95% CI 1.53 – 3.09, p<0.001 • Proteinuria HR 1.94, 95% CI 1.53 – 2.58, p<0.001		Low
Major adverse cardiovascular event (MACE) [2]	1 Retrospective cohort at outpatient office within the Cleveland Clinic Healthcare system (N=58,535) After propensity matching: *Study group* 426 patients referred to hospital *Control group:*	No significant difference in MACE between study group and control group • 7 days (0 vs 2 [0.5%]; p=0.11 • 8 to 30 days (0 vs 2 [0.5%]; p=0.11	Equivalent	Low

	852 patients sent home Follow-up of 6 months	• 6 months (8 [0.9%] vs 4 [0.9%]; p>0.99		

COST

Upon performing a thorough literature search, there were no published economic evaluations on the different sets of diagnostic work-up for hypertension-mediated organ damage in the setting of acute severe BP elevation. The components of such work-up were variable in published studies, hence full economic evaluations may be challenging.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

The term "hypertensive urgency" is contradictory to its management, which is non-urgent in nature. Due to this misnomer, providers have traditionally managed patients with acute severe hypertension with aggressive BP reduction. To date, no evidence supports this practice as it does not translate to additional CV benefit and may even cause harm. This has led to the change in terminology endorsed by this guideline and reflected by practice guidelines from other professional groups. [3]

RECOMMENDATIONS FROM OTHER GROUPS

Guidelines from the Western hemisphere, including those by the ESH, ACC/AHA, and NICE, do not recommend hospitalization for patients with acute severe hypertension and no HMOD. Instead, these guidelines recommend early follow-up and clinical re-assessment within 7 days. [4-6] Practice guidelines from Asia have mixed recommendations, with Japanese guidelines advocating against emergent treatment of patients with acute severe hypertension [7], while guidelines from China and Malaysia state that these patients may be admitted. [8,9]

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3.3 BLOOD PRESSURE LOWERING AND CARDIOVASCULAR OUTCOMES

RECOMMENDATION 3

Among adult patients with acute severe hypertension whose BP does not respond after two hours of rest, we suggest the use of oral antihypertensive medications* to decrease BP.

(certainty of evidence very low, strength of recommendation weak)

*initiation or intensification of existing oral antihypertensive medication

BEST PRACTICE STATEMENT 2

Among adult patients with acute severe hypertension, we suggest out-of-office BP monitoring with home BP or 24-hour ambulatory BP monitoring and follow-up within one week.

CONSIDERATIONS

The CP discussed the following considerations when formulating this recommendation for patients with hypertensive urgency:

- Lack of benefit in terms of BP reduction and CV outcomes when giving anti-hypertensive medications emergently (within 15-30 minutes) compared to rest of 2 hours
- Lack of evidence on the benefit/harm of giving anti-hypertensive medications emergently (within 15-30 minutes) compared to rest beyond the 2-hour timepoint

- Lack of direct evidence on the benefit/harm of giving anti-hypertensive medications emergently (within 15-30 minutes) compared to routine initiation and/or intensification of medication on outpatient basis
- Risks of aggressive BP reduction in terms of additional monitoring, medical tests, medications, and unnecessary referrals
- Need for individualizing evaluation for cause of the BP elevation, tailoring treatment, and shared decision-making when deciding to emergently treat patients with acute severe hypertension, given the heterogeneous clinical characteristics of these patients

The CP decision was unanimous regarding the positive direction of the recommendation, favoring a rest period of 2 hours prior to treatment. After one round of voting on the strength of recommendation, the CP also all agreed that the strength was weak due to the limitations of available evidence. The CP also recognized the lack of evidence beyond the 2-hour time frame; hence, a Best Practice Statement was drafted through the collaboration of the SC and the CP members to guide follow-up care. After one round of voting, the CP were unanimous in approving the Best Practice Statement.

KEY FINDINGS

- Among adults with hypertensive urgency, the administration of antihypertensive drugs within 15-30 minutes was not significantly different from two hours of rest in terms of reducing SBP and DBP at the ED. The incidence of cardiovascular mortality, stroke, and major adverse cardiac events (MACE) rates were also not significantly different at two hours.
- The administration of antihypertensive drugs within 15-30 minutes was well-tolerated and there was no increase in adverse events such as hypotension.

CHARACTERISTICS OF STUDIES

Five studies were found (1 RCT, 3 retrospective cohort, 1 retrospective descriptive) and involved a total of 24,338 patients. All studies were primarily done in Asia – one RCT in South Korea [1], two retrospective cohort studies done in Thailand [2,3], one observational study done in Thailand [4], and one restrospective cohort done in China.[5] Most studies included only patients with acute severe hypertension (formerly “hypertensive urgency”) [1-4], except for one study which had patients with hypertensive crisis not further subdivided into urgency or emergency.[5] All studies were done in the ED and assessed the effect of administering antihypertensives.

Outcomes measured included BP reduction [1,2,4], MACE including stroke, myocardial infarction, aortic dissection at 1 and 7 days [2], 7-day, 30-day, and 60-day hospital revisits, CV mortality at 1, 3, and 5 years, incidence of stroke at 1, 3, and 5 years [5], and adverse events. [2-4]

Efficacy outcomes

A single-center, randomized control trial evaluating the efficacy of resting and antihypertensive medications involving 138 patients with hypertensive urgency was conducted last 2017 by Park, et al. [1] The study investigated the effect of 2 hours of rest and antihypertensive medications on BP and the rate of BP lowering. Blood pressure measurements were repeated after 2 hours and patients were categorized by degree of BP reduction from 10 to 35% (target BP reduction) (see Table 5). They found that the number of patients who achieved a mean BP reduction of 10-35% was 68.5% in the resting group and 69.1% in the antihypertensive therapy arm, showing that there was no significant difference between the two groups (p=0.775). The mean changes in the SBP and DBP of both groups (rest vs. antihypertensive drugs) was also not significantly different (-32.2 ± 23.3 vs -32.8 ± 24.5, p=0.065 and -21.2 ± 18.5 vs -18.6 ± 17.5, p=0.352, respectively). Of the 138 participants, 99 (49 in resting group and 50 in antihypertensive therapy group) were subsequently followed up 24 hours and 7 days after emergency room admission. The SBP and DBP were also not significantly different between the two groups at 24 hours (p=0.459 and p=0.512, respectively) and at 7 days (p=0.397 and p=0.293, respectively).

The results of the prior trial were echoed in a 2020 retrospective cohort study conducted by Sricharoen et al. [2] In that study (N=692), they compared the revisit rates at 1 and 7 days, MACE (stroke, myocardial infarction, congestive heart failure, aortic dissection) at 1 and 7 days, reduction of blood pressure, and blood pressure control within 2 weeks among patients with hypertensive urgency who were treated with oral antihypertensives (n=298) and those who were not (n=394). Results showed that between the two groups, there was no significant difference in 1-day revisits [adjusted OR 0.66 (CI: 0.33-1.35), p=0.256], 7-day revisits [adjusted OR 0.99 (CI: 0.70-1.41), p=0.964], MACE [adjusted OR 0.23 (CI: 0.01-4.18), p=0.321], and BP controlled in 2 weeks [adjusted OR 0.76 (CI: 0.45-1.30), p=0.319] (see Table 5). Among the secondary outcomes in the study, the reductions of SBP and DBP were also not significantly different in those who were treated with antihypertensives and those who were not (p=0.180 and p=0.589, respectively). However, these outcomes measured only short-term risks and may not reflect the long-term risks or benefits of anti-hypertensive therapy.

A larger retrospective cohort study that involved 22, 906 adult patients, showed minimal benefit in treating patients with severe hypertension with antihypertensives. The study by Lin et al. in 2021 (N=22,906) examined patients who were treated in the emergency department and received pharmacologic BP intervention (n=6364) and those who had non-pharmacologic BP intervention (n=16,542).[5] The study showed an 11% and 11% reduction in hospital revisits at 30 days (CI: 0.82-0.97) and at 60 days (CI: 0.82-0.96) in the pharmacologic BP intervention arm. Although there was a 6% reduction in hospital revisit at 7 days among those treated with antihypertensives, it was not clinically significant [adjusted hazard ratio (aHR) 0.94 (CI: 0.84-1.06)]. There was also no clinically significant reduction in CV mortality at 1 year [aHR 0.97 (CI: 0.67-1.41)], 3 years [aHR 0.95 (CI:0.75-1.19)], and at 5 years [aHR 0.89 (CI:0.74-1.08)] among patients treated. Both groups had equivalent incidence of stroke at 1 year [aHR 0.84(CI: 0.59-1.19)], 3 years [aHR 0.84 (CI: 0.66-1.08)], and 5 years [aHR 0.81 (CI:0.65-1.01)] (see Table 5). However, the same study also demonstrated that treatment with antihypertensives during an episode of severe hypertension may be more beneficial among those with polypharmacy as evidenced by a 17% (CI: 0.75-0.93) and 22% (CI:

0.72-0.86) reduction in the risk CV mortality at 3 and 5 years, respectively. It is of note that the study did not differentiate whether the patients presented with acute severe hypertension or hypertensive emergency.

Table 7. Efficacy of antihypertensives in reducing BP and CV outcomes among patients with acute severe hypertension

Critical OUTCOMES	BASIS (No and Type of Studies, Total Participants)	EFFECT SIZE	INTERPRETATION	CERTAINTY OF EVIIDENCE
Blood pressure reduction	1 RCT (N=138) and 1 retrospective cohort (N=692)	No significant difference in achievement of mean blood pressure reduction of 10-35% and no significant difference in mean systolic and diastolic BP reduction	Equivalent	Very Low
Cardiovascular Mortality (5 years)	1 retrospective cohort (N=22,906)	aHR 0.89 (CI: 0.74 - 1.08)	No significant difference	Very low
Stroke (5 years)	1 retrospective cohort (N=22,906)	aHR 0.81 (CI: 0.65 – 1.01)	No significant difference	Very low
Myocardial Infarction (7 days)	1 retrospective cohort (N=692)	aOR 0.23 (CI: 0.01 – 4.18)	Inconclusive	Very low

Safety outcomes

A retrospective cohort study done in 2018 by Kotruchin, et al. determined the prevalence of acute severe hypertension in an ED in Thailand.[3] In the study of 451 patients who presented with acute severe hypertension, 408 were given anti-hypertensive medications and there were no noted adverse events upon administration regardless of number of antihypertensives given or combinations of antihypertensives. Similarly, in the aforementioned study conducted by Sricharoen et al., adverse outcomes from antihypertensive medications (hypotension) at 1-day and 7-day duration and found no incident of hypotension secondary to antihypertensives. [2]

Similar findings were observed in a retrospective descriptive study by Sruamsiri et al., wherein medical records of patients at the ED with severe hypertension (SBP >180 and/or DBP >120) without end organ damage (n=151) were retrieved.[4] Patients were given oral antihypertensives (captopril, hydralazine, nifedipine, or amlodipine) and repeat BP measurements were taken at 30 or 60 minutes depending on the drug given. No adverse events were noted throughout the emergency visit; however, the study did not disclose the duration of observation of these patients (see Table 6).

Despite the demonstrated tolerability of antihypertensive medications, caution should be exercised during initiation, especially with aggressive BP management due to the possible occurrence of hypotension post-discharge. [6]

Table 8. Safety of antihypertensives vs. placebo/standard of care in acute severe hypertension

OUTCOME	STUDIES	EFFECT SIZE	INTERPRETATION	CERTAINTY OF EVIDENCE
Adverse events	2 retrospective cohort (N=1143) and 1 retrospective descriptive study (N=151)	No adverse effects (e.g. hypotension) noted (N=1386)	No evidence of harm	Very low

RESOURCE IMPLICATIONS

A survey done in the Philippines (N=806) last 2023 showed that uncontrolled hypertension significantly impacts productivity especially those 40 years and above. Work-wise, 72% percent of the respondents experienced four days of missed work and 63% had 10 days of lost work. The impact is not solely limited to work but also affects day-to-day household activities and social interactions as much as 11 days of reduced household activities in 78% of respondents and 5 days of limited social activity in 60%. The overall burden attributed to direct and indirect care of uncontrolled hypertension last 2020 was at Php 52.6 billion accounting for 70% of the total economic burden and is expected to increase to Php 97.3 billion by 2050. [7]

A cross-sectional survey from the perspective of outpatients showed that the cost (synonymous to charges for this analysis) of antihypertensive medications in the Philippines is dependent on the type of medication, dose, brand, and availability and may range from Php 2.00 per tablet (losartan from the public market) to as much as Php 67.5 per tablet (lisinopril from the private market). In general, drugs procured from the private market are more expensive but are more readily available. [8] It is of note that 40% of those being treated for hypertension pay for their medications out-of-pocket and assuming that they earn minimum wage, antihypertensive medications cost 0.1 to 3.1 days' worth of wage. [7]

PATIENT'S VALUES, EQUITY AND PREFERENCE, ACCOUNTABILITY AND FEASIBILITY

Patient regard for possible outcomes such as stroke varied among the participants. In some patients, despite having a relative or even themselves experience stroke or hospitalization secondary to markedly elevated BP, adherence to their medication was still poor and appeared not to have influenced intake of antihypertensive medications, only taking them when their blood pressure was high. However, there were those who became more adherent after experiencing hospitalization secondary to complications from hypertension. Presence of the complications also urged more relatives to be more supportive of patients being adherent to their maintenance medications as they recognize the gravity of the illness as well as its financial burden. [9]

Acceptability of antihypertensive medications also varied and was affected by patient education or awareness of their medications. While patients accept that the medications are effective in lowering blood pressure, this was sometimes outweighed by their concern regarding the possibility of organ damage secondary to the medication. Other concerns expressed included the possibility of dependence on their medications which patients did not like, and the thought that having maintenance medications and focusing on adherence stressed them out which they believed worsened their hypertension. [9,10]

Given the substantial cost of antihypertensive medications, the government attempts to remedy this by providing antihypertensive medications to local barangay health centers for free. However, these centers frequently run out of medications and outpatient services are not as readily accessible to patients, making it more difficult for patients to have access to the medications. [9]

RECOMMENDATIONS FROM OTHER GROUPS

Recommendations from the US, Europe, Japan, China, and Malaysia agree that patients with acute severe hypertension can be treated with oral medications and there is no indication for emergent lowering of BP among these patients (level of evidence not stated). [11-15] Guidelines from the Japanese Society of Hypertension and European Society of Hypertension in particular advocate for gradual lowering of BP over 24 to 48 hours (level of evidence not stated).[12,13] The Malaysian Society of Hypertension guidelines states that patients with acute severe hypertension whose BP responded with 2 hours of rest may be discharged with a hypertensive urgency discharge plan, comprised of BP monitoring, medications, and follow up care (level of evidence B). For patients whose BP did not respond with 2 hours of rest, they recommend starting combination oral pharmacotherapy to target a 25% BP reduction over 24 hours (level of evidence C).[15]

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3.4 ROUTE OF ADMINISTRATION

RECOMMENDATION 4

Among adult patients with acute severe hypertension, we recommend the use of oral antihypertensive medications over sublingual antihypertensive medications to lower BP.

(certainty of evidence very low, strength of recommendation strong)

CONSIDERATIONS

The initial draft recommendation proposed that there was insufficient evidence to recommend for or against the use of sublingual or oral medications among patients requiring rapid BP reduction. In line with this, the CP members discussed the following considerations:

- Insufficient evidence on the direct comparison of sublingual and oral medications in terms of:
  - Degree of BP lowering
  - Rate of BP lowering
- Lack of benefit for lowering BP urgently, as would occur with administration of sublingual medication, in cases of acute severe hypertension
- Possible harms and adverse events associated with lowering BP urgently, such as watershed infarcts
- Shorter half-life of sublingual medications, which may predispose to greater BP variability and poorer CV outcomes
- Greater requirements for patient education when prescribing sublingual medication compared to oral medication
- Lesser complexity of the medication regimen afforded by the use of oral medications, which may promote long-term adherence

Considering the above points, the CP members were unanimous in revising the draft recommendation to be in favor of oral antihypertensive medications over sublingual antihypertensive medications in the setting of acute severe hypertension. After one round of voting, all members voted for a positive direction of recommendation in favor of oral medications. During the voting on the strength of recommendation, 6 out of 8 eligible CP members formed a consensus for a strong recommendation, while the two remaining members (MHO and PSH representatives) voted for a weak recommendation, citing lack of direct evidence for head-to-head comparison of oral and sublingual medications as the reason for their decision.

A second draft recommendation was drafted proposing the use of oral ACE inhibitors over oral calcium channel blockers, but the CP members unanimously chose to discard this draft recommendation due to its lack of correlation with the clinical question and the need for individualizing the choice of antihypertensive medication according to patient profile, which would render inappropriate a blanket recommendation favoring one drug class over another.

KEY FINDINGS

- In managing patients with acute severe hypertension, oral antihypertensives have demonstrated significant effectiveness. Lacidipine 4 mg and clonidine 0.1 mg significantly reduced SBP, while reductions in DBP were observed with lacidipine 4 mg, nicardipine 30 mg, and nifedipine 20 mg.
- No significant differences were found in the reduction of SBP and DBP between oral and sublingual antihypertensive drugs.
- Tailored treatment approaches and adherence to guideline recommendations are crucial for optimizing patient outcomes and minimizing risks associated with acute severe hypertension.

CHARACTERISTICS OF STUDIES

This review includes three systematic reviews encompassing a total of 55 randomized controlled trials (RCTs) with 2,105 participants. [1–3] Each study focuses on evaluating various antihypertensive interventions for treating hypertensive urgency in outpatient and emergency settings. The interventions reviewed include a range of medications like calcium channel blockers (CCBs), angiotensin-converting enzyme inhibitors (ACEis), and others. Campos et al. (2018) examined antihypertensive drugs for reducing BP within 48 hours while noting diverse interventions and adverse effects. [2] Souza et al. concluded that Angiotensin-converting enzyme ACE) inhibitors were more effective than CCBs with fewer side effects [3]. Cherney & Straus (2002) identified several promising antihypertensive agents for hypertensive urgencies, although inconsistent outcomes and definitions across studies were noted. [2] All three analyses emphasize the need for larger, high-quality studies with standardized methods to clarify the best treatment approaches for hypertensive urgencies and to improve long-term outcomes. In this evidence summary, we included 8 RCTs from Campos et al., 5 RCTs from Cherney & Straus and 5 studies from Souza et al. In determining the effectiveness of oral and sublingual antihypertensive drugs, were 261 participants in the treatment group and 264 participants in the control group.

RESULTS

Efficacy outcomes

We included 8 RCTs from the studies of Campos et al. (2018) and Cherney & Straus (2002). [1,2] In determining the effectiveness of oral and sublingual antihypertensive drugs, there were 261 participants in the treatment group and 264 participants in the control group. Meta-analysis was not performed due to heterogeneity of data.

Studies involving oral antihypertensives utilized oral CCBs like nifedipine and nicardipine, beta-blockers like labetalol, and alpha agonists like clonidine. [4-8] In one study, significant reductions in SBP and DBP were reported after 24 hours, with p-values of 0.008 and 0.001, respectively. Furthermore, significant reductions in both systolic and diastolic blood pressure were observed after 24 hours of treatment with both oral nifedipine 20 mg and oral lacidipine 4 mg. [4] The mean SBP decreased to a greater extent in the lacidipine group 4 mg (mean: 165 mmHg, SD: 27.3 mmHg compared to those given nifedipine 20 mg (mean: 190.6 mmHg, SD: 18.2 mmHg (p= 0.008). Similarly, the mean DBP was significantly lower in the lacidipine group (mean: 99.9 mmHg, SD: 12.3 mmHg compared to the nifedipine group (mean: 117.2 mmHg, SD: 11.4 mmHg (p= 0.001). Similarly, significantly greater reductions in DBP were observed in a inpatient study comparing oral nicardipine 30 mg to placebo (p<0.003). [5] Significant reductions in SBP were also noted in two additional studies. [7,8] Atkin et al. investigated the effects of labetalol 200 mg versus clonidine 0.1 mg at varying regimens. The study found that after 4 hours, the mean SBP was significantly lower in the clonidine group (149 mmHg, SD 7.2) compared to the labetalol group (166 mmHg, SD 4.7) (p <0.05). [7] Similarly, Jaker et al. compared the effects of nifedipine 20 mg against clonidine 0.1 mg on both SBP and DBP. After 1 hour, the mean SBP in the nifedipine group was lower at 152 mmHg (SD 6.2) compared to the clonidine group with mean SBP of 185 mmHg (SD 3.6) (p <0.05). At 1 hour, patients who received nifedipine also had a significantly lower mean DBP of 100 mmHg (SD 3.2) compared to 120 mmHg (SD 2.6) in the clonidine group (p <0.01). At the 2-hour mark, the mean DBP in the nifedipine group was 101 mmHg (SD 2.9) versus 113 mmHg (SD 2.6) in the clonidine group (p <0.05). [8] In the study of Habib 1995 et al., there was a significant DBP reduction (mean: 104.6 mmHg, SD: 13.8 mmHg among inpatients who took nicardipine 30 mg compared to the placebo group (p <0.003). [5]

Studies by Komsuoglu et al., Maleki et al., and Kaya et al. predominantly utilized sublingual calcium channel blockers (CCBs) such as nifedipine and nicardipine, along with angiotensin-converting enzyme inhibitors (ACEi) like captopril. [30–32] Across various durations ranging from 1 to 4 hours, the reported mean reductions in SBP and DBP were not statistically significant (p >0.05). However, a notable exception was observed in the study by Maleki, which demonstrated a significant reduction in SBP when taking nifedipine (5 drops) (mean: 143 mmHg, SD: NR) compared to nitroglycerin (1 pearl) (mean: 150 mmHg, SD: NR (p= 0.001). [32]

In comparing the two routes of administration, sublingual captopril 25 mg exhibited non-significant reductions in SBP and DBP in 1 hour compared to oral captopril 25 mg. [30]

In conclusion, oral antihypertensives appear to be effective in reducing BP over longer durations and in certain inpatient settings. While sublingual antihypertensives may offer rapid relief, they generally do not show statistically significant reductions when compared with other sublingual antihypertensives in the short-term durations evaluated in most studies.

Safety outcomes

Participants receiving ACEi experienced fewer adverse effects such as flushing and headache compared to those on CCB Additionally, one trial reported hospitalization due to treatment failure with CCB use and ACEi drugs. [3] Another trial noted the occurrence of angina following the administration of CCB for acute severe hypertension. These safety outcomes provide valuable insights into the adverse effects and complications associated with ACEi and CCB drugs in managing acute severe hypertension. The findings underscore the importance of considering safety profiles when selecting appropriate medications for patients with severe elevations in BP.

The relative effect (RR) for flushing with ACEi compared to CCB is 0.22, with a 95% CI ranging from 0.07 to 0.72. This finding suggests that ACEi can reduce the risk of flushing by up to 78%, with the CI indicating a potential reduction range between 28% and 93%. For headaches, the relative effect is 0.34, with a 95% CI ranging from 0.13 to 0.92, indicating that ACEi can reduce the risk of headaches by 66%, with the CI suggesting a reduction range between 8% and 87%. These results underscore that ACEi are associated with a significantly reduced risk of both flushing and headaches compared to CCBs, while accounting for the potential variability within the confidence intervals. [3]

COST

A systematic review by Park et al. sought to investigate the relative cost-effectiveness of antihypertensive medications by reviewing cost-analyses published from 1990 to August 2016. Of note, these studies were performed among patients with known chronic hypertension and not in situations of severe acute elevation of BP. Outcomes were reported as natural units, such as life year, BP reduction, or cardiovascular event avoided; utility units, such as quality-adjusted life year [QALY]); or monetary units. Majority of the studies were performed in Europe or North America and were from a healthcare system or third-party payer perspective. Results of the review showed that CCBs such as nifedipine and amlodipine were found to be more cost-effective than ACEi like lisinopril, captopril, and enalapril. Despite these findings, cost-effectiveness in the local setting may vary due to differences in the setting, methods, and perspectives of analysis. [34]

In a patient's perspective of out-of-pocket costs, the retail prices of ACEi and CCB in the Philippines can vary depending on the brand (innovator drug or brands of generic drugs) and dosage. Enalapril and lisinopril generally range from PHP 15.00 to PHP 20.00 per tablet. For instance, Ritemed enalapril 20 mg costs approximately PHP 16.00 per tablet (MIMS, RiteMED, Watsons Health Hub). In contrast, CCB like amlodipine are typically less expensive, with prices around PHP 8.00 to PHP 12.00 per tablet for a 5 mg or 10 mg dose. Overall, ACEi tend to be slightly more expensive compared to CCB in the Philippines.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

Studies in high-risk patients suggest that ACEi are superior to CCB and other drugs in protection against cardiovascular events and renal disease. Long term prospective study from the Glasgow Blood Pressure Clinic and the UK General Practice Research Database strongly support an advantage of ACEi over CCBs for cardiovascular morbidity and mortality. There should be wider use of ACEi, along with low-dose diuretics and beta blockers. [35]

Patients' preferences include satisfaction with controlled blood pressure, absence of adverse events, lower adverse event rates in ARBs, likelihood to switch from previous class (especially with ARBs), and influence of physician recommendation. Participants with blood pressure BP) controlled to JNC 7 guidelines were more satisfied with their medication than those with uncontrolled BP (90.3 vs 71.5%, $p<0.05$). Patients who had not experienced adverse events had higher satisfaction than patients experiencing adverse events (90.9 vs 75.8%, $p<0.05$). [36]

Calcium channel blockers were used most popularly with a utilization rate of 43.22%. Compound formulation and ACEI were in the second and third places with utilization rates of 38.41% and 24.97% respectively. [37]

RECOMMENDATIONS FROM OTHER GROUPS

The ESH strongly recommends against the sublingual administration of antihypertensives like nifedipine due to the risks of unpredictable and excessive BP reduction, and no recommendations were made either for or against the use of oral antihypertensive medications such as esmolol, metoprolol, and labetalol due to insufficient evidence. In the ESH guideline, CCB are the first choice for untreated patients as they have few or no contraindications and do not interfere with the diagnostic work-up for secondary hypertension. They emphasized tailoring medication choice and administration to the individual patient's needs, with careful BP and clinical status monitoring to prevent adverse outcomes like hypotension or organ damage. [33]

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3.5 RATE OF BLOOD PRESSURE LOWERING

RECOMMENDATION 5

Among adult patients with acute severe hypertension, we recommend gradual reduction of BP over 24 to 48 hours over immediate lowering to normal levels to reduce all-cause mortality, MACE, and ED re-visits.

(certainty of evidence very low, strength of recommendation strong)

CONSIDERATIONS

The initial draft recommendation proposed that there was insufficient evidence to recommend for or against gradual BP reduction over 24 to 48 hours among patients with hypertensive urgency. In line with this, the CP members discussed the following considerations:

- Lack of controlled clinical trials comparing gradual BP reduction versus rapid BP reduction among patients with hypertensive urgency
- Lack of direct evidence evaluating the 24 to 48-hour timeframe endorsed by clinical guidelines
- Lack of benefit for lowering BP rapidly among patients with acute severe hypertension in observational studies
- Very low certainty of available evidence due to risk of bias, indirectness, imprecision
- Possible harms of rapid BP lowering, such as watershed infarcts
- Influence of institutional policies, legal considerations, and patient values on the decision to rapidly lower BP in patients
- Severity of complications of markedly elevated hypertension, including conditions which may lead to disability and death

- Presence of biopsychosocial features of illness, such as the anxiety, fear, and emotional distress caused by elevated BP, which factor into the decision and timeline for BP reduction
- Need for emphasis on patient education to appease the patient's emotional response, regardless of treatment decisions
- Difficult logistics of follow-up in geographically isolated areas, which may preclude prompt follow-up visits and necessitate earlier attempts to control BP
- Key role of non-pharmacologic interventions such as lifestyle modification in overall disease management and promotion of patient empowerment
- Enabling effect of setting a timebound goal for BP reduction, enhancing provider adherence to guidelines

Considering the above points, the CP members were unanimous in revising the draft recommendation to be in favor of gradual BP reduction over 24 to 48 hours over rapid BP reduction among patients with acute severe hypertension. After one round of voting, all members voted for a positive direction of recommendation in favor of gradual BP reduction over 24 to 48 hours. During the voting on the strength of recommendation, all CP members decided on a strong recommendation, given the potential harms of rapid BP lowering coupled with lack of evidence on benefit.

KEY FINDINGS

- Observational studies on ED patients with acute severe hypertension do not provide definitive evidence of benefit or harm for withholding antihypertensive treatment. These studies did not specifically focus on gradual BP reduction; instead, outcomes were inferred from cases where antihypertensive medications were withheld and compared to those who received drug treatment.
- While non-use of antihypertensive therapy led to more gradual lowering of BP, this has not been shown to improve short- and long-term clinical outcomes.
- Withholding acute BP treatment did not result in an increase in overall ED revisits, suggesting no clear benefit for acute treatment.

RESULTS

In total, 578 articles were identified through the search process. The available evidence regarding blood pressure reduction in hypertensive urgency remains scarce. Randomized controlled trials investigating this topic are currently lacking. Two observational studies examined the outcomes of patients presenting with hypertensive urgency, comparing those who received immediate intervention in the emergency department (ED) to those who did not. These studies were motivated by the ongoing lack of consensus in the medical community regarding optimal management strategies for significantly elevated blood pressure in the ED.

A single center study by Levy et al. included 1016 patients, predominantly African American with hypertensive urgency at the emergency department, of whom 435 (42.8%) received antihypertensive therapy. The primary medication administered was oral clonidine (88.5%), with other treatments including IV labetalol (10.1%), hydralazine (7.6%), enalapril (1.6%), and metoprolol (0.5%). Those who were treated were more likely to be hypertensive, presented with elevated BP as their chief complaint, were more likely to report a headache on arrival, and had higher mean initial systolic and diastolic BP at triage (mean systolic BP (SBP) of 202 mm Hg and a mean diastolic BP (DBP) of 115 mm Hg, compared to the untreated group with a mean SBP of 185 mm Hg and DBP of 106 mm Hg). Interim BP reduction was nearly double for patients who were given an antihypertensive agent, with a drop in systolic BP of 39.8 mm Hg (vs. 19.7 mm Hg) and diastolic BP of 21.1 mm Hg (vs. 10.0 mm Hg). ED revisits at 24 hours and 30 days, hospital admission for hypertension related complications at 24 hours and 7 days were similar whether they received antihypertensive therapy or not. All-cause mortality was low with no difference at 30 days in treated and untreated groups (0.2% vs. 0.2%; difference of 0, 95% CI -1.1 to 0.8) or 1 year (2.1% vs. 1.6%; difference of -0.5, 95% CI -2.5 to 1.2). The occurrence of adverse events was minimal and death rates were low in both treated and not- treated patients. Overall, the study did not demonstrate clear benefit nor harm from not treating elevated BP within these specific timeframes. [1]

In a study conducted at Ramathibodi Hospital, a university-affiliated tertiary care facility in Bangkok, Thailand, 682 patients were seen within the Emergency Department for acute severe hypertension. Of these, 298 patients were treated with antihypertensive drugs, primarily hydralazine (58.39%) and captopril (12.42%), with 26.51% receiving combination therapies. Patients who received antihypertensive agents had a higher mean systolic BP (SBP) at presentation (205.22 ± 19.51 mm Hg) and mean diastolic BP (DBP) (101.73 ± 16.35 mm Hg) compared to those who did not receive treatment, who had a mean SBP of 194.31 ± 15.19 mm Hg and mean DBP of 94.06 ± 14.35 mm Hg. Significantly larger reduction with systolic BP of 36.38 ± 26.08 (vs 30.90±26.80) and diastolic BP of 15.34±14.93 (vs 11.33±14.39) was noted at the ED among those given antihypertensive agents. Oral hypertensive agents conferred no advantage in reducing the risk of 1-day (OR=0.58, 95% CI=0.26-1.27) and 7-day ED revisits (OR=1.11, 95% CI=0.77-1.61) with elevated blood pressure, nor did they affect ED length of stay or blood pressure control within 2 weeks. There were no reported major adverse cardiovascular events (MACE) such as Stroke, Myocardial Infarction, Congestive Heart Failure and Aortic Dissection within day 1. MACE within 7 days in those not treated and treated with antihypertensive medications was similar albeit inconclusive (OR 2.26, 95% CI 0.23-21.90). [2]

The studies did not specifically aim to gradually lower BP. Instead, conclusions were drawn based on the withholding of antihypertensive medications. Current evidence does not indicate a clear benefit or risk for not treating or treating elevated BP in the ED, suggesting no definitive advantage for either approach. This underscores the need for careful patient monitoring and individualized follow-up as potentially safe and reasonable management options in such cases. Additionally, it underscores the need for further research to provide more definitive guidance on optimal management.

It is important to recognize that these retrospective cohort studies have inherent limitations in controlling variables and establishing cause-and-effect relationship.

Selection bias was present, as patients were assigned to treatment groups based on clinical decisions rather than random assignment. The single-center nature of these studies further limited the generalizability of the findings. Additionally, data collection depended on medical records, which carries the risk of inaccuracies or missing information. Blood pressure measurements were based on chart reviews and lacked continuous 24 to 48 hour monitoring.

COST

Managing elevated BP through gradual versus immediate treatment strategies has significant economic implications. Hypertensive urgency can often be managed in the outpatient setting, which helps reduce ER visits and hospital admissions and typically relies on more affordable oral medications. However, costs can accumulate over time due to repeated monitoring and follow-up consultations. For instance, visits to a DOH Regional Hospital cost a patient ₱ 50.00 for each OPD consultation [3], adding to the financial burden.

The PhilHealth Konsulta Benefit Package enhances primary care access by covering essential antihypertensive medications and fundamental diagnostic workups. [4] This coverage supports early intervention, better chronic disease management, and improved adherence to treatment, which can reduce complications and lower the need for long-term hospitalization.

Conversely, immediate BP-lowering interventions often come with higher initial out-of-pocket costs incurred by the patient due to intensive medication use, side effect management, and potential inpatient care. In previously mentioned DOH Regional Hospital, ER visits cost a patient ₱100.00 per visit. [3] However, other public hospital emergency rooms may start at higher rates, typically beginning at ₱300 and above, offering essential care but often with high patient volumes. Variations in the charges in public or DOH hospitals demonstrate differences in the cost shouldered or funding provided either by DOH or local government units. On the other hand, private hospital ERs charge significantly more, ranging from ₱1,500 to ₱15,000. [5] Balancing these treatment approaches requires careful consideration of both short-term expenses and the long-term clinical and financial outcomes to optimize care and cost-effectiveness.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

While patient values and preferences, equity issues, and feasibility are crucial considerations in formulating recommendations for blood pressure management in hypertensive urgency, there is a notable lack of studies specifically addressing these aspects in the local setting.

RECOMMENDATIONS FROM OTHER GROUPS

Guidelines from the ESH and ACC/AHA recommend gradual lowering of BP over 24 to 48 hours. [6,7] These societies along with ACEP and ISH state that routine ED intervention is typically unnecessary for these patients and that they can be treated with oral antihypertensive medications. [8,9] As to the degree of lowering, the ESH and

Malaysian guidelines, suggest reducing BP by 25% over 24 hours, while the ACC/AHA advises reducing SBP by no more than 25% in the first hour, aiming for 160/100 mm Hg within 2–6 hours and normal BP within 24–48 hours. [6,7,10] The NSW and BIHS emphasize even more gradual reductions over days or weeks, minimizing the risk of adverse events. [11,12]

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HYPERTENSIVE EMERGENCY

3.6 RATE OF BLOOD PRESSURE LOWERING

RECOMMENDATION 6

Among adult patients with hypertensive emergency and intracerebral hemorrhage (ICH), we suggest rapid BP lowering to SBP<140 mmHg within 1 hour to prevent worsening of stroke.

(certainty of evidence low; strength of recommendation weak)

BEST PRACTICE STATEMENT 3

Among adult patients with hypertensive emergency and complications other than ICH, we suggest rapid BP control to lower SBP as indicated in the table below.

Table 7. Individualized targets and timeframes for BP lowering in hypertensive emergency

(Source: 2020 International Society of Hypertension global hypertension practice guidelines [14])

Clinical presentation	Timeframe and target BP	First-line treatment
Malignant hypertension with or without TMA or acute renal failure	Several hours, MAP -20% to -25%	Nicardipine (first line) Labetalol (if available)
Hypertensive encephalopathy	Immediate, MAP -20% to -25%	Nicardipine (first line) Labetalol (if available)
Acute ischaemic stroke and BP >220 mmHg systolic or >120 mmHg diastolic	1 h, MAP -15%	Nicardipine (first line) Labetalol (if available)
Acute ischaemic stroke with indication for thrombolytic therapy and BP>185 mmHg systolic or >110 mmHg diastolic	1 h, MAP -15%	Nicardipine (first line) Labetalol (if available)
Acute coronary event	Immediate, systolic BP <140 mmHg	Nitroglycerine (first line) Labetalol (if available)
Acute cardiogenic pulmonary oedema	Immediate, systolic BP <140 mmHg	Nitroprusside or nitroglycerine (with loop diuretic)
Acute aortic disease	Immediate, systolic BP <120 mmHg and heart rate <60 b.p.m.	Esmolol and nitroprusside or nitroglycerine or nicardipine
Eclampsia and severe pre-eclampsia/HELLP	Immediate, systolic BP < 160 mmHg and diastolic BP <105 mmHg	Labetalol or nicardipine and magnesium sulfate

BP, blood pressure; HELLP, haemolysis, elevated liver enzymes and low platelets; TMA, thrombotic microangiopathy.
Immediate: within minutes; several hours: 2 to 6 hours

CONSIDERATIONS

The following issues were discussed by the CP members as they formulated the recommendation:

- Narrow scope of available evidence covering only RCTs among patients with acute ICH, where benefit was observed only for the outcome of preventing stroke deterioration with no significant effect on other MACE such as ACS or CV mortality
- Lack of evidence for cases of hypertensive emergency with other complications aside from ICH
- Heterogeneity of clinical characteristics among patients with hypertensive emergency, which may lead to variability in response; some conditions benefit more from rapid reduction such as acute cardiogenic pulmonary edema and aortic dissection
- Need to individualize BP reduction targets for other conditions since the target of SBP <140 mmHg was tested only for patients with acute ICH
- Possible harms of rapidly lowering BP, such as loss of perfusion to the ischemia penumbra in cases of ischemic stroke
- Recommendations from other groups regarding the rapid lowering of BP within a 1-hour time frame, which were based on expert opinion
- Need for more resources accompanying the decision to rapidly lower BP, including transport to ED, additional monitoring, and medications

The above issues fueled discussion on the scope of the initial draft recommendation, which initially included all patients with hypertensive emergency. Upon appraisal of the evidence, the CP members all agreed to constrain the recommendation to patients with hypertensive emergency and ICH. After one round of voting, the CP members all agreed on the positive direction of recommendation in favor of rapid lowering of BP within 1 hour to SBP<140 mmHg for patients with ICH. Similarly, all members voted for a weak strength of recommendation due to the low certainty of evidence.

Furthermore, all the members agreed on the necessity for guidance on the management of patients with hypertensive emergency but without ICH. An existing guideline from the ISH provides a recommendation on this topic, and was appraised using the AGREE-II tool as part of the GRADE ADOLOPMENT approach. Two evidence reviewers independently appraised the guideline and found inadequate scores for the following domains: rigour of development, applicability, and editorial independence, hence the guideline's recommendation could not be adopted or adapted. Instead, the CP members drafted a best practice statement to describe guidance for cases of hypertensive emergency presenting with complications other than ICH. A representative from the PCEM stated that except for ischemic stroke, they would agree with rapid BP lowering among patients with aortic dissections, flash pulmonary edema, eclampsia, acute sympathetic crisis, and hypertensive encephalopathy. This was seconded by a representative from PHA.

All CP members agreed that the table provided in the ISH guideline [14] encapsulated the individualization in BP lowering target and timeframe that would help guide management for different cases of hypertensive emergency. With input from different members, it was decided that the table would be appended to the best practice statement after the following revisions: removal of the row on ICH as this is already covered by the recommendation, removal of columns not pertinent to the best practice statement, and clarification of the time ranges provided in the table (see above Best Practice Statement for the appended revised table). After one round of voting, all CP members agreed on the best practice statement and the appended table (see Table 8).

KEY FINDINGS

- Among patients with ICH, rapid BP lowering showed significant effect in terms of reducing neurologic deterioration, hematoma volume expansion, and adverse event. However, it did not show any significant effect in terms of reducing myocardial infarction and cardiovascular death.
- The number of serious adverse events were also significantly less among those in the rapid blood pressure lowering group.
- All studies had risk of bias issues as there were concerns in allocation concealment, blinding, and unclear detection bias. The risk of bias contributed to downgrading of evidence to low certainty of evidence for the outcome of mortality, stroke and cardiovascular death and a very low certainty of evidence for myocardial infarction.

CHARACTERISTICS OF STUDIES

The review yielded four (4) RCTs, which included a total of 10,970 patients with hypertensive emergencies. One RCT involved patients with intracerebral hemorrhage admitted in hospitals from Australia, China and South Korea [3]. The treatment group received immediate anti-hypertensive treatment based on the 1999 AHA Guidelines for the Management of Spontaneous Intracerebral Hemorrhage. They were given either Nitroprusside if initial BP>230/140, Labetalol, Esmolol, Hydralazine or Enalapril if BP>180-230/105-140. The goal is to lower SBP<140 or MAP<130. The control group meanwhile is also given anti-hypertensive drugs, but with a higher target SBP of less than 180 mm Hg. [4] The second RCT also involved patients with intracerebral hemorrhage admitted in hospitals admitted in Australia, Chile, China, France, Japan, and South Korea [5]. The treatment group is given intravenous medications to lower blood pressure (SBP<140) within 1 hour. The drugs used singly or in combination were urapidil, nicardipine, nimodipine, labetalol, nitroglycerin, furosemide, nitroprusside and hydralazine. The control group can also be given the same IV anti-hypertensives based on the 2015 AHA Guidelines for the Management of Spontaneous Intracerebral Hemorrhage, with the target SBP of less than 180. [6]

The third RCT includes patients with intracerebral hemorrhage in hospitals in study is a multi-center study with patients in USA, China, Japan, South Korea and Germany. The treatment group received IV nicardipine drip immediately, with the goal of lowering the SBP <140 mm Hg. In contrast, the control group mostly received IV nicardipine drip 3-4.5

hours after, with the target SBP<180mm Hg. [7]. The fourth study is a multi-center cluster RCT on intracerebral hemorrhage patients with severe hypertension in Australia, Brazil, Chile, China, India, Japan, Mexico, Pakistan, Peru, Sri Lanka, and Vietnam [8] The treatment group will receive IV BP lowering agents upon randomization (Nicardipine, Labetalol, Nitroprusside, Hydralazine, Urapidil) with the goal of SBP< 140 and oral medications also be commenced immediately. In addition, the group will also receive a bundle of care as adjunct such as intensive control of blood glucose with the target of 6·1–7·8 mmol/L for without diabetes and 7·8–10·0 mmol/L for patients with diabetes; treatment of pyrexia with the goal of body temperature < 37·5°C and the reversal of abnormal anticoagulation in those taking warfarin using fresh frozen plasma or prothrombin concentrate complex with the goal of reaching an international normalized ratio of less than 1·5 within 1 h of treatment. All target concentrations within the care bundle were to be maintained in patients for 7 days. The control group will only receive IV BP lowering medications if SBP>180 mm Hg.

Outcomes measures during the follow-up period of approximately 24 hours-90 days included all-cause mortality and disability [5, 7-8], ordinal analysis of functional recovery scores using the modified Rankin Scale [5, 7-8], health related quality of life using EQ-5D [5, 8], change in intracerebral hematoma [3], change in intracerebral edema [3], and adverse events [5, 7-8]. Myocardial infarction as an outcome was included in the adverse event in three trials [5, 7-8].

RESULTS

Efficacy outcomes

Based on 3 RCTs, rapid control of elevated blood pressure did not show statistically significant difference compared to conservative care in reducing all-cause mortality (RR 0.90, 95% CI 0.81-1.01). In terms of neurologic deterioration or worsening of stroke, three studies showed that rapid control of blood pressure significantly had fewer episodes than those in the control group (RR 0.89, 95% CI 0.84-0.95) (see Table 8).

In terms of other outcomes, two RCTs also measured the HRQOL using EQ-5D. There was no significant difference in terms of mobility (RR 0.86, 95% CI 0.73-1.02), self-care (RR 0.90, 95% CI 0.77-1.03) and usual activities (RR 0.91, 95% CI 0.71-1.04). Rapid blood pressure showed significant improvement in terms of pain/ discomfort (RR 0.80, 95% CI 0.67-0.93) and anxiety/ depression (RR 0.83, 95% CI 0.68-0.98).

One study measured the expansion of hematoma, and rapid blood pressure lowering showed significantly lower hematoma expansion compared to control (Mean Difference 2.8, 95% CI 1.04-4.56, p = 0.002) but no difference in terms of increase in edema volumes (Mean Difference 2.38, 95% CI -0.45 to 5.22, p = 0.10). [3] A different study showed that there is no significant difference in terms of duration of hospital stay (IQR 12-35 vs. 11-33 days, p = 0.43) [5]. Another study used proportion of patients who were discharged after 7 days, which was significantly higher in patients with rapid BP lowering (OR 0.72, 95% CI 0.53-0.98, p = 0.03) [8].

Safety outcomes

In terms of safety, three RCTs had overall adverse events as outcome, and those in the rapid blood pressure control group had statistically significant lower events (RR 0.79, 95% CI 0.73-0.85). Myocardial infarction was listed in these studies as an adverse event, and there were six cases in the treatment group and seven in the control in three studies, showing lack of statistical difference (RR 0.86, 95% CI 0.29-2.55).

Table 8. Rapid vs non-rapid blood pressure lowering of hypertensive emergency

Critical OUTCOMES	STUDIES	EFFECT SIZE	95% CI	INTERPRETATION	CERTAINTY OF EVIDENCE
All-cause Mortality	3 RCTs (n=10,704)	RR 0.90	0.81, 1.01	No difference	Low
Stroke (Deterioration)	3 RCTs (n=10,704)	RR 0.89	0.84, 0.95	Significant	Low
Myocardial Infarction	3 RCTs (n=10,704)	RR 0.86	0.29, 2.55	No difference	Very Low
Adverse Events*	4 RCTs (n=10,970)	RR 0.79	0.73, 0.85	Significant	Low

* Includes serious and fatal and non-fatal adverse events

COST

Upon performing a thorough literature search, there were no published economic analyses on rapid BP lowering compared to non-rapid blood pressure lowering in hypertensive emergencies. While it will entail additional cost, especially if parenteral therapy will be considered, it is uncertain if it will translate to less expense of the patient due to avoidance of morbidity and even additional medical interventions. The closest study done locally was a cross-sectional study using interviews and Labor Force Survey. The study mentioned that 8% of respondents were admitted due to hypertensive emergencies. Uncontrolled hypertension translated to loss of at least four days of work (72%) and ten days of work lost (63%) in majority of the subjects. The economic cost of hypertension is expected to increase from Php52.6 billion ($1 billion) in 2020 to Php97.3 billion ($1.8 billion) by 2050. [9] Hence, it may be possible that immediate control of blood pressure can lessen costs by possibly avoiding complications like worsening of stroke.

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

There are no direct studies with regards to rapid lowering of blood pressure. But based on studies, there are still considerable variations in knowledge, attitudes, and practices of Filipinos with regards to hypertension. Most consults from hypertension arises from the fact that patients have symptoms such as headache, dizziness, and dyspnea. [10] These patients would also practice some form of initial self-management such as taking an anti-hypertensive medication, herbal remedies, taking rest and relaxation, and lifestyle changes. Only then if the blood pressure is not controlled after considerable period they would go for a medical consult. [11] Other factors that motivates patients to manage their hypertension includes a) avoidance of future complications due to hypertension, b) desire to maintain blood pressure in the normal range while avoiding increase in dose of medications and c) the hope that there will be minimal impact of hypertension in their lives. [12]

It is possible therefore that majority of patients will accept and even expect immediate initial management especially if blood pressure is excessively high. Most, if not all, Filipinos would prefer an intervention, especially in an emergency room setting. For clinicians, administering medications to rapidly lower blood pressure is feasible, though evidence above still suggest lack of strong proof of efficacy. It is also unknown if rapid lowering of blood pressure leads to equity in health among Filipinos.

RECOMMENDATIONS FROM OTHER GROUPS

AHA/ACC guidelines recommend specific approaches to BP lowering in cases of hypertensive emergency. For adults with a compelling condition, which were enumerated as aortic dissection, severe pre-eclampsia, or eclampsia, or pheochromocytoma crisis, SBP should be reduced to <140 mm Hg during the first hour and to <120 mm Hg in aortic dissection. For adults without the above conditions, SBP should be reduced by no more than 25% within the first hour; then, if stable, to 160/100 mm Hg within the next 2 to 6 hours; and then cautiously to normal during the following 24 to 48 hours (Both Level I Recommendation and Quality of Evidence Expert Opinion). [13] The ISH similarly recommends that BP should be immediately controlled in the following clinical scenarios: hypertensive encephalopathy, acute ischemic stroke with SBP > 220 mmHg or DBP >120 mmHg, acute ischemic stroke with indication for thrombolysis and SBP>185 mmHg or DBP > 110 mmHg, acute coronary event, acute pulmonary edema, acute aortic disease, pre-eclampsia, and eclampsia (no level of evidence). [14] Meanwhile, the ESH recommends that BP should be immediately controlled in the following clinical scenarios: malignant hypertension with or without acute renal failure, hypertensive encephalopathy, acute coronary event, acute pulmonary edema, acute aortic disease, pre-eclampsia, and eclampsia (no level of evidence). [15] The latter two guidelines did not provide specific BP targets or timelines for the complications mentioned, Finally, guidelines from Malaysia had a Grade B recommendation for reducing BP by 10%-25% within minutes to hours but not lower than 160/100 mmHg with a preference for the use of parenteral drugs. For patients with severe pre-eclampsia, eclampsia, and pheochromocytoma crisis, a Grade C recommendation was to reduce SBP to < 140 mmHg during the first hour. For patients with aortic dissection, it was recommended to reduce SBP to < 120 mmHg. For all other complications, it was recommended to reduce BP by no more than 25% within the first hour; then, if stable, to 160/100 mmHg within the next 2 to 6 hours; and then cautiously to normal during the following 24 to 48 hours. [16]

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3.7 PHARMACOLOGIC OPTIONS

3.7.1 NITRATES

RECOMMENDATION 7

Among adult patients with hypertensive emergency and a cardiovascular event*, we recommend the use of IV nitrates to lower BP and reduce all-cause mortality.

(certainty of evidence low, strength of recommendation strong)

*acute coronary syndrome, acute cardiogenic pulmonary edema

CONSIDERATIONS

The CP members discussed the following considerations:

- Evidence on the benefit of IV nitrates for preventing CV mortality and complications, particularly among patients with acute CV events
- Low certainty of evidence due to indirectness of studies on IV nitrates, wherein the presence of CV events was the main inclusion criterion, rather than the diagnosis of hypertensive emergency
- Relatively low cost and high availability of IV nitrates compared to other IV antihypertensives
- Variable and potentially harmful effects of IV nitrates for specific patient groups with hypertensive emergency, such as patients with cerebrovascular hemorrhage

Upon discussion, the initial recommendation which proposed the use of IV nitrates among all patients with hypertensive emergency, was narrowed to pertain to only patients with hypertensive emergency and a CV event, particularly acute coronary syndrome and acute cardiogenic pulmonary edema. This was done upon consideration of findings from the best available evidence and the possible harm of IV nitrates for patients with hypertensive emergency but no CV event. During the voting for the direction of recommendation, all CP members decided on a positive direction in favor of IV nitrates for the reasons cited above. For the strength of recommendation, all members agreed that the recommendation must be strong since the benefits of IV nitrates for the specified population outweighed the risks.

KEY FINDINGS

- There is lack of direct evidence on the effect of IV nitrates in reducing BP and other CV outcomes in patients with hypertensive emergency. In hypertensive emergency, most studies were specific on the clinical outcomes of specific diseases such as acute coronary events and acute cardiogenic pulmonary edema.
- In 3 separate cohort studies [1-3], it was found out that nitrates have no significant benefit in all-cause mortality in patient presenting with acute cardiogenic pulmonary edema. It was however revealed in one of the studies that early use of

IV nitrates may improve symptoms and possibly reduce mortality, especially for patients presenting with systemic BP >180mmHg.

- Although clinical practice guidelines have recommended the use of nitroglycerin (in combination to beta blockers) as a means to reduce BP in patients with proven or suspected acute aortic disease, there is lack of direct evidence supporting beneficial outcomes. [4,5]

CHARACTERISTICS OF STUDIES

One systematic review and meta-analysis of 65 randomized control trials assessed all-cause mortality among different anti-hypertensive medications in patients with acute coronary events. In the said study, 18 trials were on nitrates. Comparator groups were given no treatment or placebo. All-cause mortality was reviewed at 2 days, 10 days and 30 days. Of the 18 eligible studies, nitrates were given as either immediate treatment (started within 24 hours of treatment and continued until 48 hours) and as short-term treatment (started within 24 hours of treatment and continued until day 10). As an immediate treatment, 6 studies reported mortality at day 2, 10 studies reported mortality at day 10, and 7 trials reported mortality at day 30. As short-term treatment, 8 trials reported all-cause mortality at day 10 and 5 trials at Day 30. Follow up period ranged from 0 to 30 days. [6]

In assessing all-cause mortality for patients with acute cardiogenic pulmonary edema, three cohort studies were retrieved and reviewed. One cohort study reviewed 8,863 Japanese patients in their acute heart failure registry (26,212 patients from January 1, 2009 to December 31, 2015), presenting with acute pulmonary edema. Outcomes measures were in-hospital mortality, length of ICU/CCU stay, and length hospital stay. Nitrates (isosorbide dinitrate and nitroglycerin) were compared with carperitide. [1] The second cohort study included 3178 individuals from the EAHFE (Epidemiology of Acute Heart Failure in Emergency Departments) registry in Spain presenting with acute heart failure. Subjects were analyzed in two groups, those who received IV nitrates and those who did not. However, only a subset of the patients (264 individuals) presented with hypertensive pulmonary edema. Clinical outcomes of mortality rates at days 3,7,14 and 30 as well as new visits at day 30 were analyzed. Follow up period for this cohort is 30 days. [2] The last study reviewed analyzed data from 947 individuals from another trial presenting with acute cardiogenic pulmonary edema. Subjects analyzed into three treatment groups (nitrates, diuretics and opiates groups). Outcomes measured were 7-day mortality, acidosis, and respiratory distress. [3]

RESULTS

Efficacy outcomes

Acute Coronary Events

In the meta-analysis, nitrates were associated with statistically significant reduction in all-cause mortality (RR 0.81, 95%CI [0.74-0.89], p<0.0001) if given as immediate treatment defined as started within 24 hours of the onset and lasting for maximum 2 days (6 trials, N=82,624). At 10 and >30 days, nitrates were not associated with statistically significant delayed reduction in all-cause mortality. As short-term treatment defined as started within 24 hours on the onset and lasting for a maximum of 10 days, six trials out of the 18 trials (N=78,178), nitrates were also associated with a statistically significant reduction in all-cause mortality (RR 0.91, 95% CI [0.86-0.97], p = 0.003. They concluded that nitrates administered within 24 hours of symptom onset significantly decrease day 2 all -cause mortality (4 to 8 deaths prevented per 1000) however no added benefit of continuation of nitrates beyond day 2. [6]

In terms of BP reduction, it was revealed that nitrates showed a statistically significant reduction in both systolic (WMD -12.67, 95% CI -14.51, -10.83) and diastolic blood pressure (WMD -7.50, 95% CI -9.07, -5.93) compared to no treatment or placebo during the first 24 hours of an acute myocardial infarction. It is, however, important to note that high blood pressure at baseline was not part of the inclusion criteria in any of the studies included in the meta-analysis. The overall weighted mean blood pressure at baseline (for the included studies) was 133/82.2 mmHg for those randomized to an active drug and 133/82.7 mmHg for those included in the placebo or no treatment group. [6] No available evidence yet on the use of IV nitrates in adult Filipinos presenting with hypertensive acute cardiovascular event.

Acute Cardiogenic Pulmonary Edema

Currently, available data however have failed to demonstrate a favorable impact of vasodilators (including nitrates) on important outcomes such as mortality. In the retrospective cohort study of Shiraishi et al. in 2019 of 8,863 (from 26,212) Japanese patients presenting with primary diagnosis of acute heart failure (presenting with pulmonary congestion on physical examination and chest X-ray findings) have showed that vasodilator use was not associated with reduced in hospital mortality (7.5% vs 8.8% in patients without vasodilators; P=0.098) as well as length of ICU/CCU (6.2 days vs 5.8 days in patients without vasodilators) and hospital stays (23.2 days vs 23.9 days in patients without vasodilators). In the same study, subset analysis showed that reductions in hospital mortality were observed with vasodilator use in patients who have SBP greater than 180mmHg (OR 0.25, 95% CI 0.10 - 0.75) as well as those who did not present with atrial fibrillation on electrocardiogram on admission OR 0.70, 95% CI 0.60 – 0.90). The reduction in mortality was not clinically significant in those who presented with severe pulmonary congestion (OR 0.79, 95 CI 0.60 – 1.02). In the same study, nitrates (nitroglycerin and isosorbide dinitrate compared to carperitide) was significantly associated with better outcomes, including in hospital mortality rates (5.1% vs 6.7% in patients with carperitide use, p = 0.041) as well as length of ICU/CCU stay (4.8 days vs 6.1 days in patients with carperitide use; p < 0.0001) and length of hospital stay (21.2 days vs 24.3 days in patients with carperitide use; p<0.0001). [1]

Another study examined the association between diuretics, nitrates and opiates in 7-day mortality among patient presenting with acute cardiogenic pulmonary edema and have observed that nitrates have lower mortality rates in comparison to the opiates and diuretics groups. However, after adjustment of confounding factors, there was no evidence of any association between treatment with nitrates, diuretics and opiates and 7-

day mortality. [3] The NITRO-EAHFE Study; a prospective, multicenter cohort study in Spain also concluded that IV nitrates do not influence early mortality or new visits in patients with acute heart failure. In the same study, a separate analysis of patient subgroup with hypertensive acute pulmonary edema has found no significant differences in mortality in between 2 groups (nitrates group vs control group), but findings may be influenced by the small number of patients analyzed in this subgroup. [2] (see Table 9)

Despite the lack of hard evidence supporting the use of IV nitrates in this subset of patients, important clinical implications may be deduced in that the early use of these drugs may improve symptoms and reduce mortality in patients with hypertensive acute pulmonary edema. [2]

There is no available evidence yet on the use of IV nitrates in adult Filipinos presenting with hypertensive acute pulmonary edema.

Acute Aortic Disease

Nitroglycerin, in combination with beta blockers, has been recommended by the European Society of Cardiology as medications to reduce blood pressure in cases of suspected aortic disease and dissection. [4] Primarily a venodilator, nitroglycerin can act as an arterial vasodilator at higher doses. [7]

Currently, there are no available studies on all-cause mortality and morbidity on the use of nitrates, specifically nitroglycerin, on acute aortic disease.

Table 9. Effect of nitrates on CV outcomes among patients with hypertensive emergency

CRITICAL OUTCOMES	BASIS	EFFECT SIZE	95% CI	INTERPRETATION	CERTAINTY OF EVIDENCE
Acute Coronary Events					
All-cause mortality	1 MT [6] (18 RCTs)	RR 0.91	0.86 to 0.97	Favors intervention	Moderate
Reduction in SBP	1 MT [6] (18 RCTs)	WMD -12.67	-14.51 to -10.83	Favors intervention	Moderate
Reduction in DBP	1 MT [6] (18 RCTs)	WMD -7.50	-9.05 t0 -5.93	Favors intervention	Moderate
Acute Cardiogenic Pulmonary Edema					
All-Cause Mortality	1 cohort [1]	OR 0.25	0.10 to 0.75	Favors intervention	Low
All-Cause Mortality	1 cohort [2]	Cannot fully estimate			Low
All-Cause Mortality	1 cohort [3]	OR 1.05	0.52 to 2.12	Inconclusive	Low

Safety outcomes
In most of the studies included in the meta -analysis of Perez et al in 2009, adverse events were not reported. Only one of the eighteen trials involving nitrates reported on adverse events related to nitroglycerine. A total of 2.6% (n = 9663) experience hypotension and 3.0% experience headache, however were not serious enough to cause withdrawal from the study. [6]

In a retrospective evaluation of 247 congestive heart failure patients presenting with acute pulmonary edema, treatment with bolus IV nitroglycerin appeared to have a favorable safety profile with asymptomatic transient hypotension (1.3%) occurring without need for further intervention.[8]

Reflex tachycardia and headache were also mentioned as side effects of nitroglycerin in the European Society of Hypertension Guidelines for the management of arterial hypertension in 2023, however, estimates were not provided. [4,5]

COST
Upon performing a thorough literature search, no economic analysis studies were found on the use of IV nitrates in hypertensive emergency compared to other IV antihypertensives. There are however a few studies suggesting potential benefit of using IV nitrates due to its relatively low cost and simplicity of dosing and administration if given as bolus. It was reported in a study that the systemic cost for a single IV NTG bottle was approximately $20. In the same study, about two thirds of the patients required only a single 1 mg dose of NTG to achieve BP lowering and improve saturation.[9] There was no available data on the economic analysis of continuous and intermittent infusion of NTG as well as IV isosorbide dinitrate.

In the Philippine, the 2021 drug price reference index (DPRI) reflecting retail prices paid by patients showed that the cost for nitroglycerin 1mg/mL, 10mL per ampoule is Php 400.00 with a range of Php 400.00 – Php 1500.00. On the other hand, the patient cost for IV isosorbide dinitrate 1mg/mL, 10mL per ampoule ranges from Php 131.78 to Php 792.00 with the DPRI of Php 131.78.

PATIENT’S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY
Currently, no data is available on Filipino adults’ values and preferences including equity and acceptability of the use of IV nitrates on the treatment of hypertensive emergency.

RECOMMENDATIONS FROM OTHER GROUPS
As mentioned earlier, the use of IV nitrates is limited to specific presenting conditions in hypertensive emergency. According to the ESH and ISH, nitroglycerine is one of the first line treatments for immediate BP lowering of patients suffering from acute coronary events, acute cardiogenic pulmonary edema (with loop diuretic) or acute aortic disease. [10]

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3.7.2 NICARDIPINE

RECOMMENDATION 8 Among adult patients with hypertensive emergency, we recommend the use of IV nicardipine to lower BP and reduce hospital length of stay. ***(certainty of evidence low, strength of recommendation strong)***

CONSIDERATIONS

The CP members discussed the following:

- Best available evidence showing that IV nicardipine lowers BP to targets faster (within 30 minutes to 1 hour) and maintains BP within target with less variability than labetalol
- Indirectness of best available evidence due to lack of controlled trials with MACE as outcomes
- Demonstrated safety of IV nicardipine even among pregnant women and populations with CKD
- Existing recommendations from other professional groups recommending IV nicardipine as first line drug for hypertensive emergency
- Widespread availability of IV nicardipine in many community and hospital settings due to inclusion of IV nicardipine in the Philippine National Drug Formulary
- High accessibility since current practices in many hospitals have IV nicardipine stocked at the ED
- Relatively affordable cost of IV nicardipine compared to other IV antihypertensives

After one round of voting on the direction of the recommendation, all CP members voted in a positive direction in favor of IV nicardipine. For the strength of recommendation, after one round of voting, all members voted for a strong recommendation due to evidence of efficacy, high availability and accessibility in many settings, and wide recognition by other professional groups as first line.

KEY FINDINGS

- Studies comparing IV nicardipine with labetalol among patients with different conditions (stroke, critical care, surgery, pediatrics, pregnancy) showed comparable efficacy and safety between nicardipine and labetalol but nicardipine had a more predictable and consistent control of blood pressure than labetalol, with less requirement of additional antihypertensive agents. [1,2]

RESULTS

Efficacy outcomes

A systematic review by Peacock et al. which included 10 studies among different populations with hypertensive emergencies and urgencies (total population n= 990, with

477 in the nicardipine group and 513 in the labetalol group) showed similar efficacy between nicardipine and labetalol. Among patients with non-traumatic ICH and a MAP of 120-160mmHg, nicardipine had a mean MAP reduction of -23.4mmHg (-15.6%) while labetalol had a mean MAP reduction of -24.1mmHg (-17.8%).[1]

One of the trials included in the systematic review evaluated hypertension in the pregnant population, there was no significant difference in nicardipine and labetalol in achieving target BP (nicardipine 70%, labetalol 63% p=0.58).[1] This was concurrent with a later network meta-analysis comparing different drugs for severe hypertension in pregnancy which showed no significant differences in the number of patients achieving target BP among nicardipine, diazoxide, nifedipine and glyceryl trinitrate. However, nicardipine required significantly more doses than nifedipine. [3,4]

Similarly, nicardipine was noted to have less variability in MAP values (nicardipine 8.19mmHg, labetalol 10.78% p=0.003), with less dosage adjustments needed (nicardipine 2, labetalol 4 p<0.001), and less need for additional agents to control the BP (nicardipine 8%, labetalol 33%, p=0.013), (nicardipine 0%, labetalol 73%, p<0.001). Moreover, nicardipine was able to achieve target BP earlier, within 1 hour among patients with ICH (nicardipine 33%, labetalol 6%, p<0.02); (nicardipine 88%, labetalol 32% p<0.001). Nicardipine also had a greater portion of time spent at goal BP in a span of 24 hours (nicardipine 82.5%, labetalol 48.5%, p<0.001), with a shorter mean time to goal BP (nicardipine 30 minutes, labetalol 90 mins, p=0.026).[1]

Furthermore, patients on nicardipine were found to have shorter hospital length of stay compared to labetalol (nicardipine 4 days, labetalol 7 days p=0.01) (nicardipine 174.6 hours, labetalol 198.5 hours, p<0.001).[1]

The findings from this 2012 systematic review were supported by a succeeding randomized control trial conducted in 2014 by Varon et al. which noted that patients administered nicardipine achieved the target range more frequently compared to those given labetalol (92% vs. 78%, P = 0.046). In terms of systolic blood pressure (SBP) measures, nicardipine-treated patients reached the target range more than those treated with labetalol (46% vs. 25%, P = 0.024). Labetalol recipients showed a higher likelihood of needing rescue medications to aid with blood pressure control (27% vs. 17%, P = 0.020).[5] (see Table 10)

A more recent systematic review and meta-analysis done in 2022, comparing nicardipine and labetalol, which included 5 retrospective cohorts and 1 prospective cohort had similar findings of nicardipine's superiority over labetalol in terms of time at goal BP among stroke patients (CI: 0.112-0.438 p=0.001). However, nicardipine was found to have higher adverse events, including hypotension and tachycardia (CI: 1.077–2.113 p= 0.017). However, the studies included a risk for selection, performance and detection bias.[6]

Table 10. Efficacy outcomes of nicardipine vs labetalol among patients with hypertensive emergency

Outcomes	Basis	Number of Participants	Relative Effect	Interpretation	Quality of Evidence
Reached target BP in 30 mins	1 RCT	104	92% vs 78%, p=0.046	Favors nicardipine	Moderate
Reached target BP in 30 mins	1 RCT	226	91.7% vs 82.5%, p=0.039	Favors nicardipine	Moderate
Achieved target BP in 1 hr	1 Retrospective study	90	33% vs 6%, p<0.02	Favors nicardipine	Low
Achieved target BP in 24hr	1 RCT	47	100% vs 68%; P<0.001	Favors nicardipine	Moderate
Achieved target BP in 1 hr	1 RCT	47	88% vs 32% P<0.001	Favors nicardipine	Moderate
Mean MAP variability	1 Retrospective study	90	8.19 vs 10.78; P=0.003	Favors nicardipine	Low
Mean dose adjustments	1 Retrospective study	90	2 vs 4; P<0.001	Favors nicardipine	Low
Need for additional agents	1 Retrospective study	90	8% vs 33%; P=0.013	Favors nicardipine	Low
Time spent at goal BP	1 RCT	47	82.5% vs 48.5%; P<0.001	Favors nicardipine	Moderate
Time to goal BP	1 RCT	47	30 min vs 90min; P=0.026	Favors nicardipine	Moderate
Percent needing additional agents	1 RCT	47	0% vs 73%; P<0.001	Favors nicardipine	Moderate
BP measurements within target range	1 RCT	226	47.3% vs 32.8%; P=0.026	Favors nicardipine	Moderate
Required second agent	1 Retrospective Study	318	34% vs 59%; P<0.001	Favors nicardipine	Low
Time spent at goal BP	1 Meta-analysis (5 retrospective, 1 prospective)	3443	SMD: 0.275 CI: 0.112-0.438 P= 0.001	Favors nicardipine	Low
Hospital length of stay	1 Retrospective Study	318	174.6h vs 198.5h P<0.001	Favors nicardipine	Low

In an RCT by Yang et al. comparing nicardipine and nitroprusside for ED patients with hypertensive crisis and acute pulmonary edema, both sets of participants saw notable drops in their systolic and diastolic blood pressure post-treatment, with no substantial differences observed over time between the groups.[7]

A more recent study by Seifi et al. in 2023, comparing clevidipine with nicardipine, showed no statistically significant difference between the two drugs, however, clevidipine reached the target SBP 23 minutes faster than nicardipine, with less total volume infused to achieve target BP.[8] As of the time of this writing, clevidipine is not available locally, limiting its applicability in the local setting.

Safety Outcomes
Both nicardipine and labetalol are safe in hypertension in pregnancy, with no significant difference in number of patients achieving target blood pressure (nicardipine 70%, labetalol 63%, p=0.58), with same time to reach target BP (nicardipine 11.1 minutes, labetalol 12.4 minutes, not significant).[1]

Subgroup analyses of the CLUE clinical trial revealed that for subjects with renal dysfunction (defined as a creatinine clearance of <75 ml/min), patients who received nicardipine were within target range more than the labetalol group, and was less likely to require rescue medications.[1]

In another study by Kim et al. in 2012 aiming to assess the safety of nicardipine among 52 study participants, complications noted during nicardipine infusion included hypotension in 1 (1.9%) and findings of tachycardia or an arrhythmia in 4 (7.7%). No patient was noted to have bleeding or renal dysfunction. [9]

COST
Upon performing a thorough literature search, no economic analysis studies were found on the use of IV nicardipine for hypertensive emergency compared to other IV antihypertensives. Nicardipine is available in the Philippines in 1mg/ml formulation 10-ml ampules. Retail prices paid by patients range from 106.23 to 420.00 pesos, while the 2021 Philippine Drug Price Index pegs it at 190.12 pesos. Given the initial dose of 5mg/hr, the local cost of the intervention incurred by the patient is equivalent to about 95 pesos per hour or 2,280 pesos per day.[10]

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY
No studies were found on patient preference towards nicardipine compared to other IV anti-hypertensives.

RECOMMENDATION FROM OTHER GROUPS
In October 2020, the American Journal of Hypertension released an article on Hypertension Management in Emergency Departments noted that the most common medications indicated for treatment of hypertensive emergencies are nicardipine, labetalol, clevidipine, and esmolol. Although nitroprusside was the mainstay for decades, the concern for cyanide toxicity made nicardipine and clevidipine the drugs of choice.[11]

The 2020 International Society of Hypertension Global Hypertension Practice Guidelines also recommends labetalol and nicardipine as first line options for hypertensive emergencies, particularly malignant hypertension, hypertensive encephalopathy, acute ischemic stroke with or without thrombolytic therapy hemorrhagic stroke and aortic diseases.[12]

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3.7.3 LABETALOL

RECOMMENDATION 9

Among adult patients with hypertensive emergency, we suggest not to use IV labetalol for reducing BP.

(certainty of evidence very low, strength of recommendation weak)

CONSIDERATIONS

The CP members discussed the following issues:

- Lack of controlled clinical trials, with only nonrandomized studies exploring the comparison of IV labetalol and IV nicardipine for patients with acute stroke
- Very low certainty of best available evidence due to indirectness, risk of bias, and inconsistency
- Lesser degree and slower rate of BP reduction with use of IV labetalol compared to IV nicardipine, with no significant difference in terms of CV outcomes
- Lower incidence of unspecified tachycardia with IV labetalol compared to IV nicardipine
- Limited availability of labetalol in most community and hospital settings and non-inclusion of IV labetalol in the Philippine National Drug Formulary
- Greater cost of IV labetalol compared to other IV antihypertensives, which may lead to challenges in procurement

During the first round of voting for the direction of recommendation, all CP members voted for a negative direction, opposing the use of IV labetalol. For the strength of recommendation, CP members unanimously decided on a weak recommendation due to the very low certainty of evidence, the inferiority of IV labetalol to IV nicardipine in existing studies among stroke patients, and barriers to availability and procurement.

KEY FINDINGS

- In a meta-analysis of 6 non-randomized controlled trials on the management of hypertensive emergency in adults with acute stroke, time to reach target BP was significantly faster for IV nicardipine when compared to IV labetalol. This superior time to reach goal BP has not been shown to directly affect cardiovascular morbidity and mortality.
- Although there was a higher incidence of tachycardia for the IV nicardipine group, this was inconsistent and both medications were even found to have comparable adverse event profile in other included studies.

CHARACTERISTICS OF STUDIES

The review yielded as systematic review of six retrospective/prospective cohort studies evaluating the safety and efficacy of IV labetalol compared to IV nicardipine in the management of acute stroke, including intracerebral hemorrhage, acute ischemic stroke, and subarachnoid hemorrhage.[6] There were a total of 3443 patients in this study, with 3219 in the IV Nicardipine group and 224 in the IV Labetalol group. Standardized mean difference (SMD) was computed to compare the mean time to reach goal BP in both groups.

RESULTS

Efficacy outcomes in acute stroke

The primary endpoint, which was time to reach target BP, was significantly better for IV nicardipine (0.275 SMD, 95 CI, 0.112-0.438, P = 0.001). However, the endpoint of superior time to reach goal BP has not been shown to directly affect cardiovascular morbidity and mortality among acute stroke patients. [1] (see Table 11)

Safety outcomes

There was a higher incidence of adverse events for the nicardipine group (OR 1.509, 95% CI, 1.077-2.113, I^2=0.00%, P=0.017) in the meta-analysis by Zhang et al. [1] Primary adverse events recorded amongst the six studies were hypotension, bradycardia, and tachycardia. It was only the rate of tachycardia in patients given Nicardipine that was found to be significantly higher. (see Table 11)

Table 11. Efficacy and safety outcomes of IV labetalol compared to IV nicardipine among patients with acute stroke

Critical Outcomes	Basis	Effect Size	95% CI	Interpretation	Certainty of Evidence
Time at goal BP lowering	5 retrospective cohort, 1 prospective cohort (N=3443)	Average difference (SMD): 0.275 SD	0.112 to 0.438	Does not favor IV Labetalol	Very Low
Adverse Event (Tachycardia)	5 retrospective cohort, 1 prospective cohort (N=3443)	OR 1.509	1.077 to 2.113	Favors use of IV Labetalol	Very low

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

There is no data on Filipino patient's value and preferences in the acceptability of the use of IV Labetalol in the treatment of hypertensive emergency.

RECOMMENDATIONS FROM OTHER GROUPS

The recommendation of labetalol as either first line or alternative therapy for hypertensive encephalopathy, ischemic stroke, intracerebral hemorrhage, acute coronary syndrome, acute aortic dissection, eclampsia or preeclampsia is similar for ISH, ACC/AHA, and ESH. [2-4] The 2020 CPG for the management of hypertension, which includes recommendations for management of acute stroke and preeclampsia/eclampsia, recommends labetalol as second choice to IV nicardipine.[5]

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3.7.4 HYDRALAZINE

RECOMMENDATION 10

Among non-pregnant adult patients with hypertensive emergency, we suggest not to use IV hydralazine for reducing BP.

(certainty of evidence low, strength of recommendation weak)

CONSIDERATIONS

The CP members discussed the following issues:

- Lack of randomized controlled trials for use of IV hydralazine among non-pregnant adults
- Low-certainty observational studies showing lack of efficacy for reducing BP or hospital LOS, with greater incidence of AEs
- Recommendations from other professional groups relegating IV hydralazine as last-line drug for resistant hypertension due to the reasons above
- Relatively affordable cost of IV hydralazine relative to other IV antihypertensives
- Limited availability of other more efficacious IV antihypertensives in geographically isolated areas

Upon the first round of voting on the direction of the recommendation, the CP members' votes diverged with 5 eligible members voting in favor of hydralazine while the other 4 members voted against hydralazine. The members voting in favor of recommending hydralazine cited the potential unavailability of other IV antihypertensives, such as nicardipine, and the inhibiting effect of a negative recommendation on providers' use of hydralazine as an alternative agent. On the other hand, the members against the use of hydralazine cited the lack of evidence for the benefit of hydralazine, with observational studies even showing harm. During the second vote, majority of the eligible CP members (7 members) voted against the use of hydralazine, while the other 2 members, comprised of the representative from PMA and one representative from PAFP, retained their positive stance due to the reasons stated above. The CP members then voted on the strength of recommendation, and all members voted for a weak recommendation due to the lack of controlled trials on this topic.

KEY FINDINGS

- There is lack of evidence for the impact of IV hydralazine on CV outcomes (all-cause mortality, stroke, myocardial infarction) among patients with hypertensive emergency. There were no randomized controlled studies, meta-analyses, or systematic reviews that looked directly at the BP reduction and major adverse cardiovascular events (MACE) in patients given IV hydralazine in the emergency room/hospital setting.
- Hydralazine is not recommended as first line IV therapy in the treatment of hypertensive emergencies due to highly variable BP changes and hypotensive episodes. The overall certainty of evidence is very low due to the observational research design of the included studies.

RESULTS

There were no studies that directly looked at BP reduction and MACE in patients with hypertensive emergencies in the hospital setting given IV hydralazine. We found 1 prospective observational study and 2 retrospective cohort studies.

A prospective observational study conducted in 2011 (n = 94) showed that IV hydralazine given for identified hospitalized patients failed to achieve a >25% reduction in BP after the initial 2-hour treatment period. Only 2% (n = 4) of patients had evidence of hypertensive emergency defined as documented hypertension-mediated organ injury or symptoms consistent with severe hypertension. The mean BP prior to administration of IV hydralazine was 175/82 ± 25/16 mmHg. During the 2 hours following IV hydralazine administration, the BP was reduced by a mean of 24/9 ± 29/15 mmHg. More marked changes were observed in those with the highest range (≥ 190 mmHg) of pre-treatment SBP (−35 ± 25 mmHg). The majority of patients (83%) who received IV hydralazine experienced less than the recommended 25% reduction in BP for the treatment of hypertensive emergency. [1]

A retrospective review in 2010 identified 2,189 patients for whom 742 orders for hydralazine were written (10-20 mg/per dose) and 5,915 for labetalol (10-20 mg/dose). Ordered drugs were administered in 60.3% of patients and the average number of doses administered was 5.3 ± 8.2 (mean + SD) for hydralazine. Hospital length of stay (LOS) for patients for whom hydralazine was ordered was 12.0 ± 15.9 days for those who received at least one dose and 7.1± 9.0 days those who did not receive a dose (p<0.001). [2]

A cohort study assessed patients who received a dose of either labetalol or hydralazine in the emergency department to determine appropriateness of IV antihypertensive use. Treatment was defined as inappropriate if documented suspicion for an indicated cardiovascular condition or acute end-organ injury was lacking. 357 patients over an 18-month period were included with mean age of 55 (51% male and 93% black). 127 (36.4%) were considered inappropriately treated, IV hydralazine given in 34% of patients. There was no difference in mean (SD) BP post-treatment at 30 or 60 minutes between groups. BP in the appropriate group was 178/100 (31/21) mmHg compared to 176/98 (25/17) mmHg in the inappropriate group at 30 minutes post-treatment (p=0.54). At 60 minutes, mean (SD) BP of the appropriate group was 177/97 (28/21) mmHg compared to 172/97 (26/16) mmHg in the inappropriate group (p=0.19). There was no statistical difference in in-hospital mortality between patients treated appropriately (2%) versus those treated inappropriately (0%). [3]

Indications for use of IV hydralazine differed in all studies. In the first, IV hydralazine was given for patients with elevated BP regardless of level of BP elevation. In the second study, definition of appropriate use of IV antihypertensives included hypertensive emergency defined as presence of hypertension-mediated organ damage or symptoms of elevated BP. [1,2] Only one study identified the IV hydralazine dose of

11.4 ± 4.3 mg which was only given in 4 patients (2%) fulfilling criteria for acute severe hypertension. [1]

Table 12. Efficacy outcomes of hydralazine among patients with hypertensive emergency

Outcomes	No of Studies Population	Result	Interpretation	Certainty of Evidence
BP reduction	1 prospective observational study from Connecticut[5] with IV hydralazine EMR order for hospitalized patients N=94 (no control) IV hydralazine given in 4 patients (2%) fulfilling criteria for hypertensive urgency Change in blood pressure 2 hours post administration and adverse effects	83% of patients who received hydralazine failed to achieve a > 25% reduction in BP after the intial 2-hour treatment period 17 patients experienced an adverse event (N 11 hypotension)	Does not favor IV hydralazine	Low
Hospital length of stay (LOS)	1 retrospective review in Michigan[6] of in-hospital EMR orders for low-dose IV hydralazine and labetalol PRN order N 2189 (no control) (742 orders for IV hydralazine 10-20 mg) control those with order but not administered	LOS was longer in patients in whom IV antihypertensive was administered vs those ordered and not given LOS for patients for whom hydralazine was ordered was 12.0 $\pm$ 15.9 days for those who received at least one dose and 7.1$\pm$ 9.0 days those who did not receive a dose (p<0.001).	Does not favor IV hydralazine	Low
In-hospital mortality	1 retrospective cohort study from Michigan[7] reviewed records in the ED with elevated BP given one dose IV anti-hypertensive therapy (IV labetalol, enalaprilat, hydralazine or metoprolol) N 357 (no control) 64 patients received IV hydralazine (39 appropriate, 25 inappropriate)	No difference in in-hospital mortality (2% vs 0%) or 30-day ED revisit rate between those given appropriate IV antihypertensive therapy (IV hydralazine Hypotension in 2 patients requiring pressor support treated inappropriately and 1 patient treated appropriately not needing pressors	Does not favor IV hydralazine	Low

Efficacy outcomes

In the prospective observational study of 94 patients (mean age 69 ± 18 years, 48% women, 89% with chronic hypertension identified hospitalized patients with an order for IV hydralazine. Only 4 (2%) of patients had evidence of an urgent hypertensive condition. Changes from baseline in BP were related to baseline BP: change in BP for those with lower baseline BP was −3 ± 20 mmHg while change observed among those with higher baseline BP was −35 ± 25 mmHg. Eighty-three percent of patients who received hydralazine failed to achieve a > 25% reduction in BP after the initial 2-hour treatment period. [1]

In the study by Miller et al. where 357 patients in the ED received at least one bolus of IV antihypertensive: either IV labetalol, enalaprilat, esmolol or hydralazine. Treatment was defined as inappropriate if documented suspicion for an indicated cardiovascular condition or acute end-organ injury was lacking. 39 patients received IV hydralazine under the definition of appropriate use and 25 patients received IV hydralazine without indication for its use. There was no statistical difference in in-hospital mortality among those patients who were appropriately treated (2%) compared to those inappropriately treated (0%). [3] (see Table 12)

Safety outcomes

Hydralazine is inconsistent in its effects and difficult to titrate. Unexpected, sudden drops in BP beyond intended targets limit its use as a treatment in the emergency room and hospital setting. [2,4] The resultant tachycardia from IV hydralazine also limits its use in patients as seen in a retrospective cohort of patients among critically ill individuals. 4.4% (n=246) experienced increases in heart rate of more than 20 bpm.[5]

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

No local evidence was found regarding impact of IV hydralazine on health equity. Although IV hydralazine is readily available in many hospitals in the Philippines, hesitation for its use due to adverse effects is weighed against questionable benefit.

RECOMMENDATIONS FROM OTHER GROUPS

The use of IV hydralazine in the treatment of hypertension in adults is no longer recommended in international guidelines for the treatment of hypertensive emergency. [6-9] In the ESC guidelines, hydralazine is now virtually abandoned because of their association with serious adverse effects. It may be occasionally considered for cases of resistant hypertension that are unresponsive to multiple agents, but not as first line therapy (strong recommendation). Among pregnant women, it has also been shown to be associated with more perinatal adverse effects. [6-8]

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3.8 PRIMARY CARE CONSIDERATIONS

RECOMMENDATION 11

Among adult patients with hypertensive emergency in the primary care setting while the patient is for transfer, we suggest the use of antihypertensive medications to reduce BP (IV preferred but if not available, oral).

(certainty of evidence very low, strength of recommendation weak)

BEST PRACTICE STATEMENT 4

Among adults with hypertensive emergency in the primary care setting while the patient is for transfer, we suggest referral to a health facility capable of monitoring and safely lowering BP.

CONSIDERATIONS

The CP members discussed the following issues:

- Very low certainty of the best available evidence on the use of antihypertensives for hypertensive emergency in the primary care setting due to indirectness, imprecision, and nonreporting of effect size in some studies
- Lack of controlled trials performed in the primary care setting among patients with hypertensive emergency who are for transfer to the ED
- Lack of recommendations from other guidelines regarding use of antihypertensives in the primacy care setting as bridge to ED management
- Heterogeneity of clinical characteristics among patients with hypertensive emergency, which may lead to variability in response
- Importance of rapid treatment for hypertensive emergency due to life-threatening complications
- Difficulties with establishing IV access and availability of IV antihypertensives while in the primary care setting
- Prevailing practices in the community, where providers often given oral antihypertensives while waiting for transfer to the ED

- Challenges surrounding the logistics of transferring patients to the nearest hospital ED, especially in geographically isolated and disadvantaged areas
- Importance of establishing linkages with the public health care provider network and local referral system to facilitate transfer
- Need for close monitoring and individualization when patients are treated at primary care setting, necessitating referral to health facilities capable of monitoring and safely lowering BP

The initial draft recommendation was in favor of administering antihypertensives to patients with hypertensive emergency who are awaiting transfer to the ED. Upon discussion of the above points, the CP members decided to include a qualifier on the type of antihypertensive medication to recommend. Given the benefits of rapid BP lowering and the potential danger of prolonging untreated hypertensive emergency, the preference for IV antihypertensives over oral hypertensives was added to the recommendation. Furthermore, recognizing the limited availability and accessibility of IV antihypertensives in isolated areas, the qualifier was formulated to allow for the use of oral antihypertensives should IV medications be unavailable. Once the recommendation was revised, the CP underwent one round of voting where all members voted in the positive direction in favor of treating patients with hypertensive emergency as they wait for transfer to the ED. After one round of voting on the strength of recommendation, all eligible CP members voted for a strong recommendation due to the importance of rapid treatment for hypertensive emergency, life-threatening complications of untreated hypertensive emergency, and logistical challenges for transfer to ED in distant areas.

Stemming from the above discussion, the topic of qualifying the type of facility for referral of patients with hypertensive emergency was raised by a representative from PAFP. CP members agreed that the facility must be capable of adequately monitoring BP, administering IV antihypertensives, and responding to any adverse events. Hence, a best practice statement on this issue was formulated. Upon one round of voting, all CP members agreed to approve the best practice statement.

KEY FINDINGS

- Overall, data from observational studies does not support that the use of antihypertensive medication resulted in significant BP reduction and improved cardiovascular outcomes after 12 months in those patients with severe elevation of BP in the primary care setting.
- The level of evidence for the review was low due to the risk of bias, observational nature of the studies, inconsistencies, and indirectness.

RESULTS

The review yielded no RCTs or meta-analyses which directly evaluated the use of antihypertensive medications in hypertensive emergency in the primary care setting. One prospective observational study assessed the therapeutic approach of general practitioners (GP) for patients presenting with severely elevated BP in the primary care setting. [1] A 2008 Cochrane review evaluated treatment outcomes with any first-line anti-hypertensive drug class compared to placebo or a different anti-hypertensive drug in participants with hypertensive emergency. [2]

CHARACTERISTICS OF STUDIES

This observational study by Merlo et al. evaluated GPs management strategy of 164 patients with severely elevated BP. Most or 99 (65%) of the patients were asymptomatic, 50 (31%) had acute severe hypertension, and only 15 (9%) had hypertensive emergency. The mean initial SBP and DBP of patients with hypertensive emergency were 211 +/- 14 mmHg and 106 +/- 15 mmHg, respectively. The outcomes measured were decline in measured blood pressure at specific time points, and cardiovascular events at 12 months. [1]

In the Cochrane study, 15 RCTs from 1983 to 2004 were included with a total of 869 participants. Although blood pressure entry criteria differed between included trials, all participants presented with acute end organ damage. The anti-hypertensive drugs used were nitrates (9 trials), ACEi (7 trials), calcium channel blockers (CCBs) (6 trials), peripheral alpha-1 blockers (4 trials), diuretics (3 trials), direct vasodilators (2 trials), and dopamine agonist (1 trial). The primary outcomes measured were total serious adverse events, all-cause mortality, and a composite of non-fatal cardiovascular events. The secondary outcomes measured include changes in systolic blood pressure (SBP), diastolic blood pressure (DBP), heart rate (HR), and withdrawals due to adverse effects (AEs). [2]

Efficacy outcomes

All-cause mortality

For the primary outcome of all-cause mortality, the review by Perez et al. was not able to pool the results of included studies. [2] There were 7 trials that evaluated mortality but only 3 RCTs reported actual deaths, which totaled 6, while the 4 RCTs reported no deaths in their study with a follow-up that ranged from 6 to 24 hours.

Composite non-fatal cardiovascular events

The review was also unable to pool the outcome of composite non-fatal cardiovascular events but reported the rates of MI and pulmonary edema requiring mechanical ventilation in five studies. One placebo-controlled trial and three other antihypertensive drugs head-to-head trials showed there was no statistical difference in MI rates between these interventions (see Table 13). Similarly, there was no statistically significant difference between placebo and captopril (RR 0.40, 95% CI 0.09 to 1.86), nitrates and alpha-adrenergic antagonist (RR 4.12 95% CI 0.20-84.24), or nitrates and ACE-i (RR 0.33, 95% CI 0.01 to 7.78).[2] Similarly, Merlo et al. found out that there was no difference in the occurrence of cardiovascular events in patients who were asymptomatic, in hypertensive urgency, or in hypertensive emergency during presentation. [1]

BP lowering

BP readings were conducted at one, six, and twelfth hour, after 2±1 days, and after 3±1 months after initial consultation. [1] Around two thirds or 63.2% of all participants received an antihypertensive medication, the most common of which was a calcium antagonist (nifedipine 20 mg). There were 37% participants who were given combination therapy. A higher proportion of participants with acute severe hypertension and hypertensive emergency received prompt BP lowering medication over asymptomatic patients (OR 7.9, CI 95% 3.4 to 18.2 p<0.0001). However, there was no difference in the practice of GP of giving BP lowering agent between participants with acute severe hypertension and hypertensive emergency. Outcomes were available for 140 (85.4%) patients after 12 months on follow-up.

Weighted mean change in BP

Drug vs placebo or no treatment

Based on the result of one study, the pooled effect of comparing three sublingual antihypertensive medications, CCB, ACEi, and alpha-1 adrenergic antagonist, to placebo showed a significant reduction in both systolic (WMD -13.14, 95% CI -19.48 to -6.80) and diastolic (WMD -8.03, 95% CI -12.61 to -3.45) BP. [2]

Nitrates vs other antihypertensive medications

The results of the pooled analysis from three trials showed that there was no statistical difference in lowering systolic (WMD -5.62, 95% CI -13.26 to 2.02) or diastolic BP (WMD -3.36, 95% CI -8.7 to 1.98) between nitrates and diuretics. This study used furosemide as the diuretic and used nitroglycerin or isosorbide as nitrate. Two studies, one using nitroprusside and the second one using nitroglycerin, compared a nitrate's BP lowering effect to an alpha-1 antagonist, urapidil. There was also no statistical difference between the systolic (WMD -3.49, 95% CI -9.31 to 1.43) and diastolic (WMD 0.76, 95% CI -2.7 to 4.23) BP reduction between these groups. [2]

In the trial comparing nitrates with a dopamine agonist, there was statistically greater reduction in systolic BP (WMD -14.00, 95% CI -27.72 to 0.28) but not in diastolic BP (WMD -2.00, 95% -11.74 to 7.74).
There was also no statistically significant BP reduction between nitrates and ACEi (SBP WMD 3.33, 95% -7.37 to 14.03, diastolic BP WMD -0.33, 95% CI -6.69 to 6.03), nitrates and CCB (SBP WMD 4.67, 95% CI -3.97 to 13.32, diastolic BP WMD 3.5, 95% CI -3.94 to 10.94), and nitrates and a direct vasodilator, hydralazine (SBP WMD -3.67, 95% CI -20.41 to 13.07, diastolic BP WMD 3.34, 95% CI -4.82 to 11.5). [2]

ACEi vs other antihypertensive medications

The pooled data from four trials comparing ACEi with a CCB showed a significant DBP reduction (WMD 7.86, 95% CI 4.92 to 10.81) but no significant reduction in SBP (WMD -1.68, 95% CI 4.92 to 10.81). ACEi was found to statistically lower SBP (WMD -20, 95% CI -22.85 to -17.39) and DBP (WMD -3.70, 95% CI -7.08 to -0.31) compared to alpha-1 adrenergic antagonist. [2]

Safety outcomes

The trial comparing alpha-blocker with nitroglycerine was the only trial that reported withdrawal due to adverse events (5% vs 2.7%; [RR 3.38, 95%CI 0.17-68.84]) which was found to be not significant. [2]

Table 13. Efficacy outcomes of giving antihypertensives for patients with hypertensive emergency

Critical Outcomes	Basis	Effect Size	95% CI	Interpretation	Certainty of Evidence
Cardiovascular event (at 12 months follow up)	1 prospective observational study (164 participants)	Effect size not given but found to be not significant (p value=0.17)	Not given	Does not favor antihypertensive medications	Very low
Acute MI					
ACE-i vs placebo	1 study (48 participants) (Hamilton 1996)	RR 0.72	0.31 to 1.72	Does not favor antihypertensive medication	Very low
Nitrates vs furosemide	1 study (69 participants) (Beltrame 1998)	RR 1.30	0.40 to 4.19	Does not favor nitrates	Very low
Nitrate vs urapidil	1 study (112 participants)	RR 0.55	0.09 to 3.17	Does not favor nitrates	Very low
Captopril vs Nifedipine	1 study (20 participants)	RR 0.5	0.05 to 4.67	Does not favor nifedipine	Low
Diazoxide vs hydralazine	1 study (52 participants) (DANISHI II 1986)	RR 0.86	0.06 to 12.98	Does not favor diazoxide	Low
BP lowering	1 prospective observational study (164 participants)	OR 7.9	3.4 to 18.2	Does not favor antihypertensive medications	Very Low
SBP reduction					
CCB or ACE-i or A1A vs placebo	1 study (118 participants) (Pastorelli 19991)	WMD -13.14	-19.48 to 0.68	Favors antihypertensive medications	Moderate
Nitrates vs diuretics	3 studies (121 participants) (Beltrame, Nelson, Verma)	WMD -5.62	-13.26 to 2.02	Does not favor	Very low
Nitrates vs A1A	2 studies (193 participants) (Hirschl, Schreiber)	WMD -3.94	-9.31 to 1.43	Does not favor	Very low
Nitrates vs Dopamine agonist	1 study (28 participants) (Elliott)	WMD -14	-27.72 to (-0.28)	Favors nitrates	Low
Nitrates vs ACE-i	1 study (46 participants) (Hirschl)	WMD 3.33	-7.37 to 14.03	Not favor	Low
Nitrates vs CCB	2 studies (100 participants) Rubio-G, Yang	WMD 4.67	-3.97 to 13.32	Not favor	Very low
Nitrates vs Direct vasodilator	1 study (24 participants) Verma	WMD -3.67	-20.41 to 13.07	Not favor	Low
ACE-i vs CCB	4 studies (183 participants) (Angeli, Marigliano, Wu, Pastorelli)	WMD 7.86	4.92 to 10.81	Favors ACE-i	Low

ACE-I vs A1A	2 studies (121 participants) Wu, Pastorelli	WMD -20.12	-22.85 to -17.39	Favors ACE-i	Low
Diazoxide vs Hydralazine	1 study (52 participants) DANISH II	WMD 13.56	3.06 to 24.06	Does not favor	Low
DBP reduction					
CCB or ACE-i or A1A vs placebo	1 study (118 participants) (Pastorelli 19991)	WMD -8.03	-12.61 to 3.45	Favors antihypertensive medications	Moderate
Nitrates vs diuretics	3 studies (121 participants) (Beltrame, Nelson, Verma)	WMD -3.36	-8.7 to 1.98	Does not favor	Very low
Nitrates vs A1A	2 studies (193 participants) (Hirschl, Schreiber)	WMD 0.76	-2.7 to 4.23	Does not favor	Low
Nitrates vs Dopamine agonist	1 study (28 participants) (Elliott)	WMD -2	-11.74 to 7.74	Does not favor	Low
Nitrates vs ACE-i	1 study (46 participants) (Hirschl)	WMD -0.33	-6.69 to 6.03	Does not favor	Very low
Nitrates vs CCB	2 studies (100 participants) Rubio-G, Yang	WMD 0.58	-2.44 to 3.59	Does not favor	Very low
Nitrates vs Direct vasodilator	1 study (24 participants) Verma	WMD 3.34	-4.82 to 11.5	Does not favor	Low
ACE-i vs CCB	4 studies (183 participants) (Angeli, Marigliano, Wu, Pastorelli)	WMD -15.79	-20 to -11.59	Favors ACE-i	Low
ACE-I vs A1A	2 studies (121 participants) Wu, Pastorelli	WMD -3.7	-7.08 to -0.31	Favors ACE-i	Low
Diazoxide vs Hydralazine	1 study (52 participants) DANISH II	WMD 14.67	8.01 to 21.33	Favors Hydralazine	Low

PATIENT'S VALUES AND PREFERENCE, EQUITY, ACCEPTABILITY, AND FEASIBILITY

There is no yield for search on studies on Filipino patient's value and preferences in the acceptability of the use of any antihypertensive medication in the treatment of hypertensive emergency in the primary care setting.

RECOMMENDATIONS FROM OTHER GROUPS

Current guidelines have no recommendation on the specific type of BP lowering agent preferred for use in the primary care setting or while awaiting transfer to a hospital for patients with hypertensive emergency. The general recommendation available advocates a BP reduction strategy performed in a setting where BP monitoring and titration of medications can be safely conducted. Common to these guidelines is careful BP reduction using easily-titratable IV antihypertensive agents, the choice of which will depend on the specific manifestation of end-organ damage in the patient. [3-5] JNC7 suggests to individualize oral antihypertensive medications to lower blood pressure in the primary care setting. [6]

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4. LIMITATIONS AND RESEARCH IMPLICATIONS

The strength of recommendations in this guideline was often limited by the lack of direct evidence on specific aspects of severe BP elevation management. For majority of the medications assessed, the focus of existing trials was on the degree and rate of BP reduction with less emphasis on the impact of the medications on CV outcomes such as MI, stroke, or MACE. Furthermore, there were limited head-to-head comparisons among different medications, precluding any decisions on the relative efficacy of one agent compared to another. The impact of different treatment approaches on CV outcomes among patients with severe BP elevation remains an active research gap, which requires the attention of international and local researchers.

On the topic of workflows, there was limited to no literature on the impact of specific referral pathways on the outcomes of patients with severe BP elevation, whether for hypertensive emergency or otherwise. Due to this, critical recommendations on referral approaches and management while in transition from primary care to ED were based on expert opinion only. The dearth of literature in the implementation aspect of these management approaches should be the topic of further investigation.

Majority of the evidence reviewed was among a Western population and most studies performed in Asia were performed among East Asians. Differences in pharmacodynamics, pharmacokinetics, and epigenetics among Asian populations may give rise to variability in BP response and outcomes, hence additional studies among Filipinos may be required to generate direct evidence. Furthermore, evidence on patient-centered outcomes, such patient-reported symptom measures and quality of life, is also limited for patients with severe BP elevation, necessitating further research.

Finally, there was lack of evidence on the local cost-effectiveness and acceptability of the various management approaches and medications used for severe BP elevation. Information on the cost-effectiveness of treatment options and acceptability according to Filipinos' values would be valuable in guiding more contextualized policies and recommendation. Hence, research on these topics should be pursued contingent with studies on efficacy and safety.

5. DISSEMINATION, IMPLEMENTATION, & ANTICIPATED IMPACT

A full copy of this document will be sent to the Department of Health for transmittal and publication. The Disease Prevention and Control Bureau will transmit copies of this CPG to the Philippine Health Insurance Corporation (PHIC) and health maintenance organizations (HMOs) and NGOs involved in the care of patients with hypertension.

The DOH planned to develop a simplified version of this CPG and made it available in the format that will be ready for reproduction and dissemination to the patients in different health care settings. It will also be available for interested parties who might visit the DOH website.

Anticipated impacts of implementation include: (1) enhanced knowledge of the evidence-based approach to and management of severe BP elevation among target users of the guideline, (2) more informed decisions by policy-makers in terms of policies and regulations affecting public programs targeting hypertension, (3) greater interest among local researchers regarding gaps in the management of severe BP elevation among Filipinos, (4) improvements in the rate of diagnosis of new cases of hypertension and rate of control among patients with known hypertension, and (5) reduction in adverse events resulting from gaps in diagnosis and management of severe BP elevation among Filipinos.

6. APPLICABILITY ISSUES

The Hypertension Task Force calls on the reader to consider equity, applicability, and individualization when applying the recommendations of this CPG in medical practice. Comprehensive history taking, physical examination, and judicious monitoring are fundamental to patient assessment and the insights gathered from these core processes should guide the overall management.

Barriers to the implementation of this guideline include: variations in the settings and populations on which it will be applied compared to those in the evidence, limited financial capacity at the level of patients and providers, differences in availability, accessibility, and affordability of the diagnostics and interventions, and the need for additional capacity-building among providers at all levels of care regarding new paradigms for acute severe hypertension.

This CPG is only a guideline and does not replace careful consideration of the patient's goals, values, and circumstances. Although this CPG intends to influence the direction of health policies for the benefit of the general public, it should not be the sole basis for revising or abolishing existing health practices.

7. UPDATING OF THE GUIDELINES

The recommendations presented in this document shall hold until such time that new evidence emerges on the topic, dictating the update of this CPG. This guideline will be updated after 3 years.

Appendix 1. Summary of Search Terms

Clinical Questions	Search Terms
History and PE	[hypertension] AND [hypertension emergency] AND [hypertensive urgency] AND [signs] AND [symptoms] AND [diagnosis] AND [accuracy] AND [sensitivity] AND [specificity] #MESH Terms
Indications for Referral	[hypertensive emergency] AND [hypertensive urgency] AND [referral] AND [referral] AND [cardiovascular morbidity] AND [mortality}
Gradual lowering	{hypertensive urgency] AND [BP lowering] AND [BP management] AND [standard of care] AND [immediate treatment] AND [cardiovascular mortality] AND [stroke] AND [myocardial infarction]
Route of administration	[hypertensive emergency] OR [hypertensive urgency] OR [high blood pressure] OR [severe asymptomatic hypertension] OR [sublingual] OR [medications]
Hypertensive emergency management	[hypertensive emergency] AND [pharmacologic management] AND [sublingual] AND [intravenous] AND [route of administration]
Nicardipine	[hypertension] AND [emergency] AND [nicardipine] AND [cardiovascular mortality] AND [cardiovascular morbidity] RCTs
Labetalol	[hypertension] AND [emergency] AND [labetalol] AND [cardiovascular mortality] AND [cardiovascular morbidity] RCTs
Hydralazine	[hypertension] AND [emergency] AND [hydralazine] AND [cardiovascular mortality] AND [cardiovascular morbidity] RCTs
Nitroglycerine	[hypertension] AND [emergency] AND [nitroglycerine] AND [cardiovascular mortality] AND [cardiovascular morbidity] RCTs

Appendix 2. Conflict of Interest Assessment

A - Allowed – participation with no constraints;

B - Broadcast - Manageable with **minor** constraints – there are COIs: participation is allowed with minor constraints (eg – COIs declared aloud during panel discussions);

C - Manageable with **major** constraints - there are COIs: participation is allowed with major restrictions (eg – disallowing voting);

D- Disallowed – participation is not allowed

Names & Positions	Assessment	Management
STEERING COMMITTEE		
Deborah Ignacio Ona	B	Changed from D to B - To declare all COI which includes the ff: - PLAS current president - PSH past president (2023) - Chair, Scientific Committee and Organizing Committee for the Joint Convention of the Philippine Society of Hypertension and Philippine Lipid and Atherosclerosis Society, (2021) - TRB - Phil. Guidelines for HPN" - Speakers bureau for the the ff: i. Servier ii. Viatris iii. Sanofi iv. Natrapharm
Gilbert Vilela, MD	B	Changed from D to B - To declare all COI which includes the ff: - Speakers bureau for the the ff: i. Novartis ii. Astra iii. Boehringer Ingelheim
Felix Eduardo Punzalan, MD	B	Changed from D to B - To declare all COI which includes the ff: - Research grant – Astra Zeneca - Speakers bureau for the ff:

		i. Astra Zeneca ii. Servier iii. Merck
Aurelia Leus, MD	B	To declare all COI which includes the ff: - Past PHA President - CPG for Hypertension
Richard Santos, MD (PCEM)	B	To declare all COI which includes the ff: - Associate Editor of Phil Journal of Emergency Med. - Speakers bureau for the the ff: i.Glaxo (GSK) ii.Ambica iii.Unilab
Noel Espallardo, MD (PAFP)	B	To declare all COI which includes the ff: i.Medical Affairs Consultant - BSV BioScience Phils., ii.Project Manager - PAFP UHC Pilot Test Pfizer

Names & Positions	Assessment	Management
TECHNICAL WORKING GROUP		
TECHNICAL LEAD		
Raymond Oliva, MD	B	Changed from D to B - To declare all COI which includes the ff: i. Medical Director Astellas Pharma ii. CPG for Hypertension iii. Speakers bureau for the the ff: PSH - hypertension lectures Natrapharm - Telmisartan (October 2024)
EVIDENCE REVIEW EXPERTS		
Katrina Mata, MD (Qx. No 2)	A	Allowed participation (no COI with the question/topic she was assigned to)
Rommel Bataclan, MD (Qx. No 5)	A	Allowed participation (no COI with the question/topic he was assigned to)

Valentin Dones, MD (Qx no 3)	A	Allowed participation (no COI with the question/topic he was assigned to)
Karl Murillo, MD (Qx no 6 and 1)	A	Allowed participation (no COI with the question/topic he was assigned to)
Hannah Almenario ,MD (Qx no 8 & 1)	A	Allowed participation (no COI with the question/topic she was assigned to)
Sam Fusingan ,MD (Qx no 7)	A	Allowed participation (no COI with the question/topic she was assigned to)
Elmer Llanes, MD (Qx no 10)	A	Allowed participation (no COI with the question/topic he was assigned to)
Pauline F. Convocar, MD (Qx no 11)	B	To declare all COI which includes the ff: i. PCEM Chair on Advocacy, ii.Advocacy on EMS
Lourdes Ella Santos, MD (Qx no 9)	B	To declare all COI which includes the ff: i. PLAS past president (2022) ii. Speakers bureau for the ff: bayer – rivaroxaban Sanofi - pcsk9 inhibitor Menarini - nebivolol Viatris - atorvastatin eplerenone Novartis - small intertering rna (pcsk9) Servier - rosuvastatin/ezetimibe, perindopril/amlodpine indapamide amgen-zuellig - pcsk9 inhibitor Organon
Cristina San Jose, MD (Qx no 4)	B	To declare all COI which includes the ff: i. lectures on Acute Stroke Management which includes BP management ii. Speakers bureau for the ff: Noacs and cilostazol
TECHNICAL WRITER		

Ma Sergia Fatima Sucaldito, MD	A	Allowed participation
CP MEETING FACILITATOR		
Diana Tamondong - Lachica, MD	B	To declare all COI which includes the ff: i.Head of the Program for Healthcare Quality and Patient Safety (UPCM) ii.Overall Coordinator of the Quality Improvement and Patient Safety in the Department of Medicine (IM-QuIPS)

Names & Positions	Assessment	Management
CONSENSUS PANEL		
Krizia Tesorio Mier, MD (PAFP)	A	Allowed participation
Minerva Calimag, MD (PMA)	A	Allowed participation
Daystar Sedillo, MD (MHO)	A	Allowed participation
Romulo Babasa, MD (PCEM)	B	To declare: Resource Speaker - Point of Care Ultrasound (GE)
Dave Gamboa, MD (PCEM)	A for qxs 1-9 & 11 C for qx no. 10	Allowed participation for qxs 1-9 & 11 Cannot vote for qx no 10: Medical Director- Julius Quiambao Medical & wellness center
Jan Cabasag (PNA)	A for qxs 1-9 & 11 C for qx no. 10	Allowed participation for qxs 1-9 & 11 Cannot vote for qx no 10: SLMC Marketing Director
Suzanne Langcauon, MD (PAFP)	B for qxs 1 – 9 & 11 C for qx no. 10	To declare: i.Resource Person, Medical Emergencies in the Workplace Cannot vote for question no 10: 1. PROSER HEALTH SERVICES, INC - Family Med Medical Coordinator

		2. Philcare Heyphil Digimed 3. Stoutcon Emergency Response
Alberto Atilano, MD (PSH)	B for qxs 1 – 9 & 11 C for qx no. 10	To declare: i. Director- Foundation for Lay Education on Heart Diseases2005- present, ii. Co-Author- Chapter on Hpn- Phil Book of Internal Medicine- 1st edition Cannot vote for qx no 10: Medical Director- Perpetual Succor Hospital and Maternity Inc. 2010- present
Denver Tiongson ,MD (PHA)	B for qxs 1, 4, 5 to 10 C for therapeutic questions (2, 3, & 11)	To declare: i. Authorship on hypertension treatment Cannot vote for qxs nos. 2, 3 and 11) Speakers bureau for the ff: Natra - telmisartan Sandoz - telmisartan Sanofi - irbesartan Menarini - nebivolol Viatris - amlodipine Novartis - entresto Gx - telmisartan Getz - bisoprolol Servier - perindopril, indapamide, amlodipine
Karen de Leon ,MD (PHA)	B for qxs 1, 4, 5 to 9 C for therapeutic questions (2, 3, 10 & 11)	To declare: i. Philippine Heart Association, HPN Council member 2023 to present Cannot vote for qxs nos. 2, 3, 10 and 11 1. Speakers bureau for the ff: Getz Pharma – nebivolol (2023) Emcure – telmisartan (2023) Novartis – Entresto (2023) 2. Stocks - Laguna Holy Family Hospital (Sta. Cruz, Laguna) and Global Care Medical Center of Bay (Bay, Laguna)
Louis Villanueva, MD (PHA)	B for qxs 1, 4, 5 to 9	To declare: i. Hypertension lectures

	C for therapeutic questions (2, 3, 10 & 11)	Cannot vote for qxs nos. 2, 3, 10 and 11) 1. Speakers bureau for the ff: Servier Menarini Astra Zeneca Corbridge Torrent Pharma Camber Pharma LRI Theraphrma Pfizer Zydus 2. Stocks - Calamba Doctors' Hospital, Unihealth Sta Rosa, Unihealth Southwoods
Ramoncito Herrera (Patient advocate)	A for qxs 1, 4, 5 to 10 C for therapeutic questions (2, 3, & 11)	Cannot vote for qxs nos. 2, 3 and 11) Provides training for the ff pharma companies: Pascual, Unilab and Sanofi
ADMIN OFFICER		
Princess Marie T. Sulit, RN	A	Allowed participation (no COI with the question/topic she was assigned to)

8. SUMMARY ALGORITHM